# Design Baseline

Named input case · design_baseline.json

2026-11-09T22:55:45+00:00

**Dry rover mass**

**43.0 kg**

Before adding regolith

**Payload per cycle**

**123.9 kg**

Total carried by four drums

**Complete cycles**

**2**

Within 15 minutes; no partial delivery credit

**Delivered volume**

**148.7 L**

Complete-cycle mission estimate

## Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

**Construction points**

**1,990.53**

Weighted mass + energy points

**Mass points**

**1,507.64**

Dry-mass productivity component

**Energy points**

**482.90**

Consumed-energy productivity component

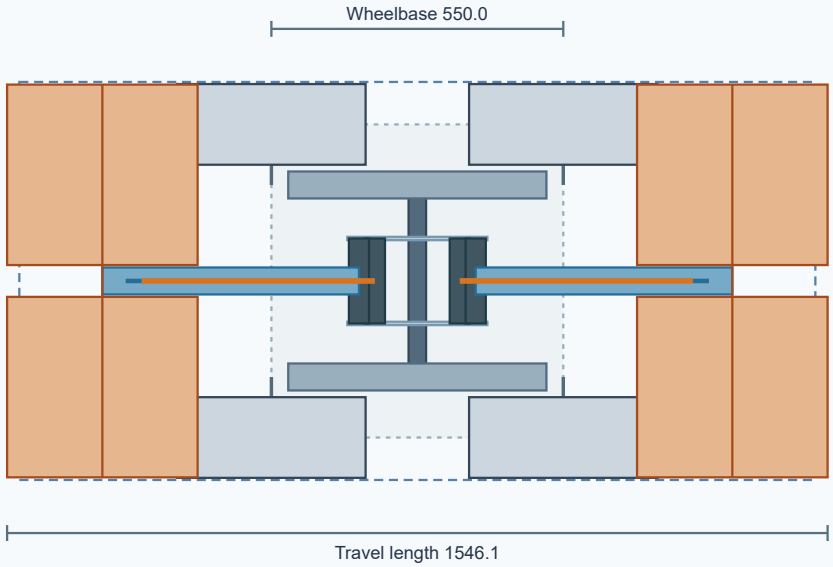
01 / ROVER CONFIGURATION

# Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

## Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.




Wheel-center track: 590.0 | Overall width: 742.0 | Wheel width: 152.0

Each drum: 359.0 diameter x 340.0 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: aluminum cross-tubes · slate: rockers and axle.

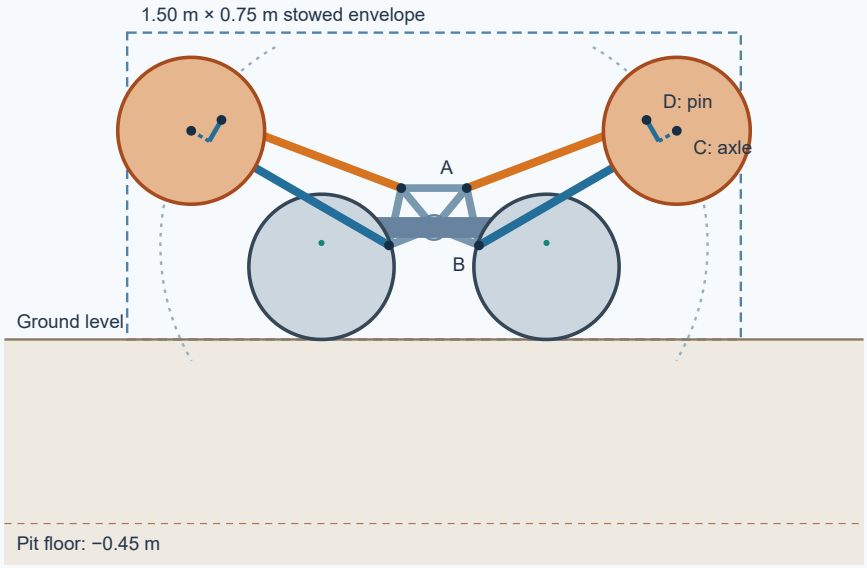
Four aluminum cross-tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

## Travel

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



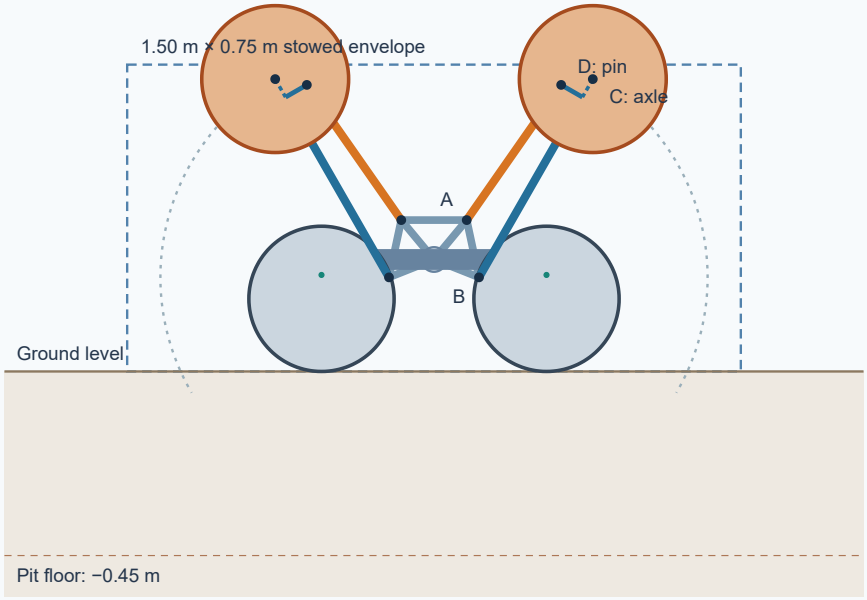

Actuator: 470.0 mm · Fixed arm: 558.8 mm · Drum Ø 359.0 mm

Drum top: +0.690 m · Drum bottom: +0.331 m · A: 0.370 m · B: 0.230 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -50.8 mm / top +60.0 mm · Ground is y = 0

## Deposition · both pairs

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



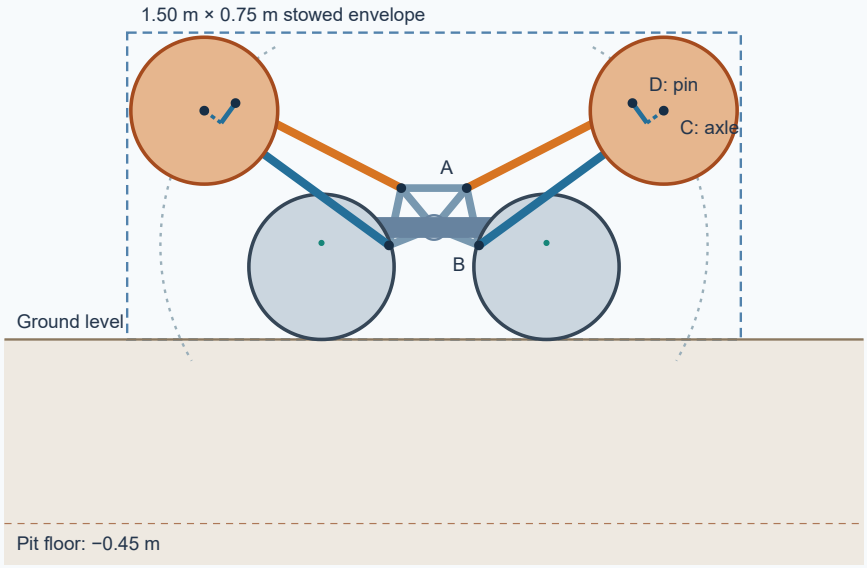

Actuator: 403.1 mm · Fixed arm: 558.8 mm · Drum Ø 359.0 mm

Drum top: +0.894 m · Drum bottom: +0.535 m · A: 0.370 m · B: 0.230 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -50.8 mm / top +60.0 mm · Ground is y = 0

## Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



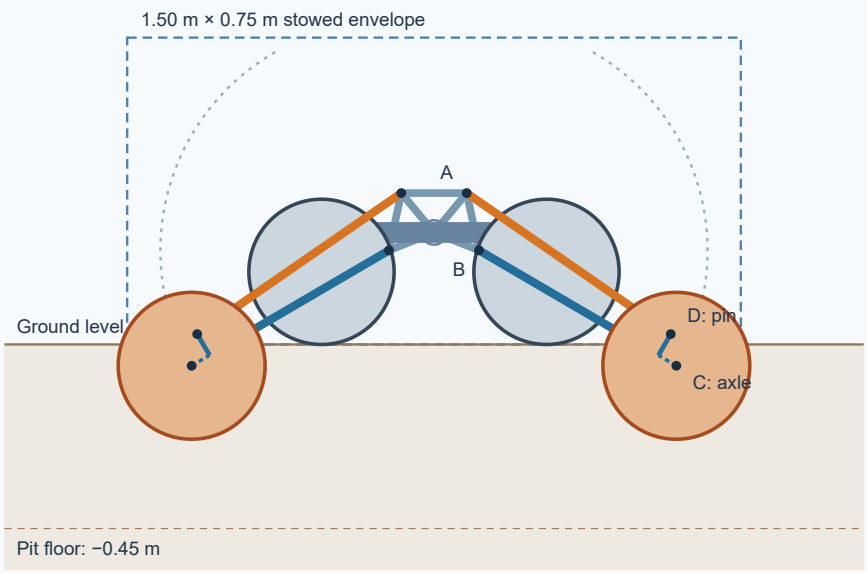

Actuator: 455.3 mm · Fixed arm: 558.8 mm · Drum Ø 359.0 mm

Drum top: +0.739 m · Drum bottom: +0.380 m · A: 0.370 m · B: 0.230 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -50.8 mm / top +60.0 mm · Ground is y = 0

## Maximum extension · plunge / recovery reach

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair




Actuator: 606.3 mm · Fixed arm: 558.8 mm · Drum Ø 359.0 mm

Drum top: +0.127 m · Drum bottom: -0.232 m · A: 0.370 m · B: 0.230 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -50.8 mm / top +60.0 mm · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

## Travel and deposition mass properties

| Pose / loading | Longitudinal CG | CG height | Yaw inertia |
|---|---|---|---|
| Travel · fully loaded | 0.00 mm | 429.0 mm | 59.4 kg·m$^2$ |
| Deposition · fully loaded | 0.00 mm | 601.2 mm | 31.2 kg·m$^2$ |
| Deposition turn · one pair full | 229.0 mm | 569.6 mm | 18.3 kg·m$^2$ |
| Deposition · empty | -1.03e-14 mm | 446.9 mm | 5.40 kg·m$^2$ |

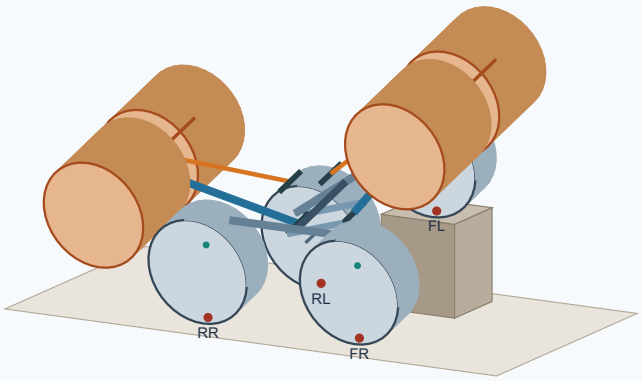
Added pose-change duration: 13.3 s. Added electrical energy: 4,201 J. Both pairs move together while full before dumping, then return empty after both dumps.

Raising reduces yaw inertia only when the drums move inward. Higher CG can reduce dynamic stability. One wheelbase must satisfy both travel and deposition checks. Half-loaded arm repositioning and dynamic dumping require controller and hardware validation.

## One wheel on the rock / travel pose

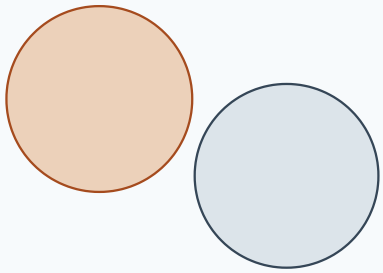
Front-left raised: 300.0 mm · chassis pitch 17.0° · roll 13.1°

Same kinematic pose as the model; arms held in travel position.




### Common roll removed

Closest cylinder pair: drum 4 / RR



Radial circle gap: +34.00 mm

Axial end gap: -151.00 mm

Finite-cylinder gap: +34.00 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: aluminum cross-tubes · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

# Plate-frame member schematic

## Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.

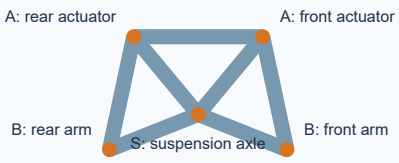



Plate thickness: 6.35 mm (fixed)

Common strip width: 18.75 mm

cross-tube ID / OD: 31.75 / 38.00 mm

Plate spacing / cross-tube span: 160.0 mm

Rocker bearing spacing: 361.8 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 50.80 x 50.80 mm

Arm wall: 1.5875 mm

Central support: pin, balanced mirrored gravity only.

cross-tube material properties are PROVISIONAL.

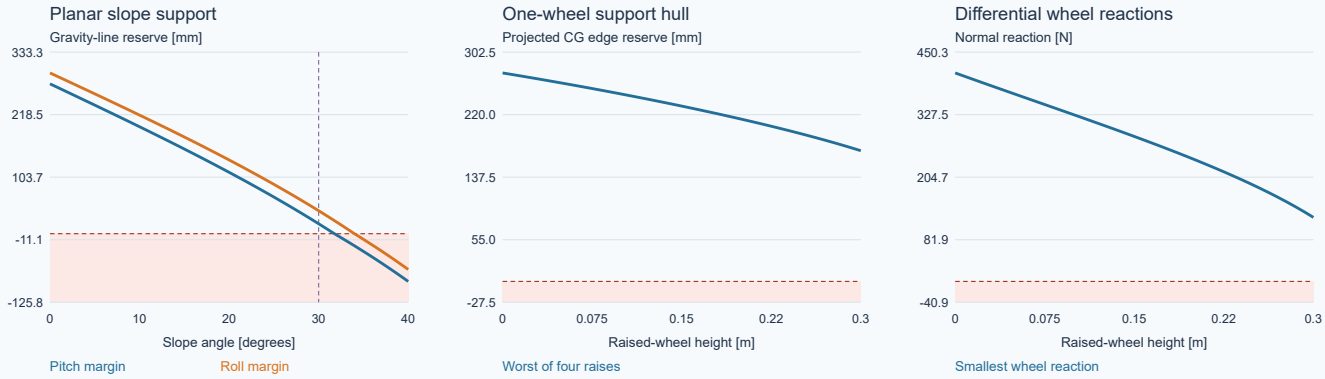
Spoke buckling is a strip screen,

not whole-plate buckling validation.

## Static slope and articulation curves

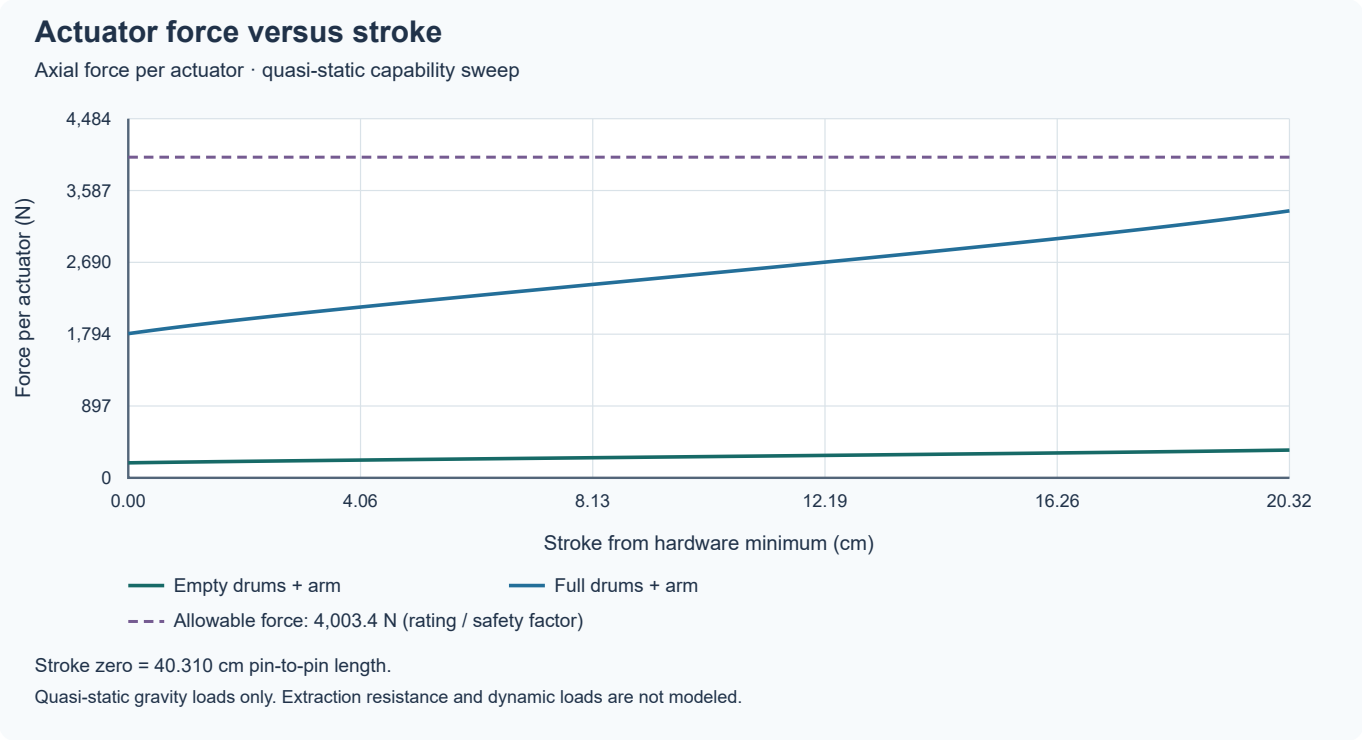
### Loaded travel pose: gravity stability screens

Linear springs + hard stops. Curves after wheel unloading are guarded continuations; slope and rock are separate cases.



LINKAGE / ACTUATOR LOADS

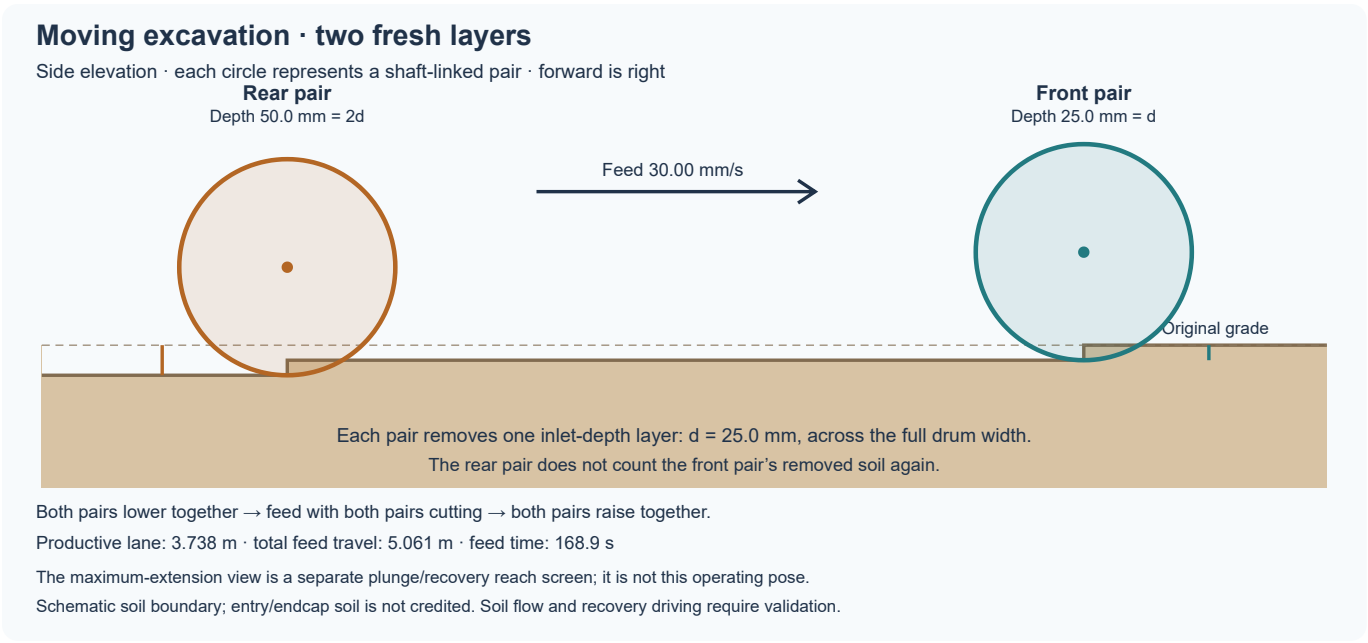
# Actuator force versus stroke



EXCAVATION / FORWARD FEED

# Drive straight with the drums lowered

Each pair cuts one inlet-depth fresh layer across its full drum width. Radial protrusion equals inlet depth; the rear cuts a second layer in the same tracks. Forward advance per scoop remains limited by inlet depth.




Both pairs lower together → feed with both pairs cutting → both pairs raise together.

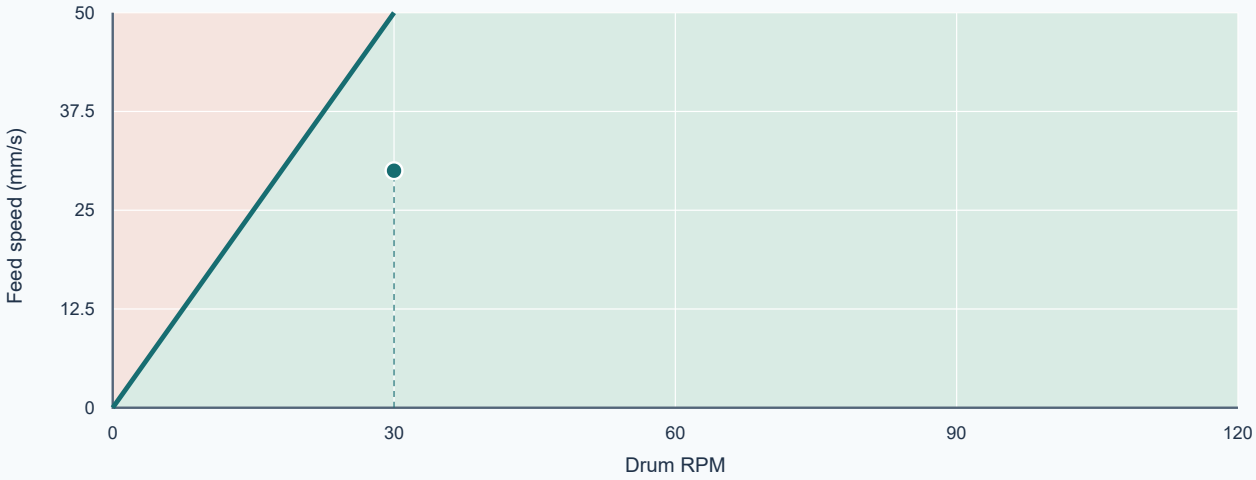
Productive lane: 3.738 m · total feed travel: 5.061 m · feed time: 168.9 s

The maximum-extension view is a separate plunge/recovery reach screen; it is not this operating pose.

Schematic soil boundary; entry/endcap soil is not credited. Soil flow and recovery driving require validation.

## Feed speed and drum RPM

Forward advance per scoop ≤ inlet depth · an uncalibrated feed screen




Current: 30.00 rpm, 30.00 mm/s

Forward advance: 15.00 mm/scoop · inlet depth: 25.00 mm

Maximum feed = inlet_depth × scoops/revolution × RPM / 60; at this RPM: 50.00 mm/s.

Green: passes this feed screen. Red: excessive advance per scoop. Motor, reach and traction checks still apply.

This volume/bite constraint does not establish measured capture, cutting loads or high-speed soil retention.

| Excavation quantity | Evaluated value |
|---|---|
| Inlet depth / fresh-layer cut depth | 25.0 mm |
| Inlet width (full drum width) | 340.0 mm |
| Front depth below original grade | 25.0 mm |
| Rear depth below original grade | 50.0 mm |
| Straight excavation feed speed | 30.0 mm/s |
| Excavation drum RPM | 30.0 rpm |
| Advance per scoop | 15.0 mm |
| Feed limit at selected drum RPM | 50.0 mm/s |
| Productive excavation lane length | 3,738 mm |
| Feed travel including tandem staging | 5,061 mm |
| Straight-feed duration | 168.9 s |
| Feed-drive electrical energy | 1,641 J |
| Feed-drive peak electrical power | 13.5 W |

| Excavation quantity | Evaluated value |
|---|---|
| Plunge/recovery: axle reaches ground level | 52.1 mm |
| Lower both arms together | 9.82 s |
| Raise both arms together | 9.50 s |
| Total excavation arm motion | 19.3 s |
| Excavation arm electrical energy · both actuators | 5,932 J |

Feed-drive electrical work and peak power are included in digging totals. The old stationary-plunge soil-volume calculation is inactive. Maximum extension remains a separate capability envelope. Reaching the ground with an axle does not verify motor clearance or driving on the drums.

02 / PREDICTED OUTPUT

# Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

| Prediction | Value | Meaning |
|---|---|---|
| Full cycle duration | 302.4 s | Includes digging, actuation, travel, dumping and relocation; overlap removed once. |
| Excavation including arm motion | 196.3 s | Both arms lower together, feed/cut, then both raise together. |
| Deposition including arm motion and turn | 27.7 s | Position both arms, dump, turn, dump, and return arms to travel. |
| Excavation + deposition | 223.9 s | Excludes route, fresh-site relocation and alignment overhead. |
| Energy per cycle | 15.4 Wh | Electrical energy including auxiliaries. |
| Mission energy · complete cycles | 32.8 Wh | Includes idle electronics for remaining mission time. |
| Mission energy · continuous estimate | 45.8 Wh | Used by the existing battery constraint and MDO objective. |
| Peak electrical power | 679.5 W | Largest modeled simultaneous demand; not an average power rating. |
| Loaded mass | 167.0 kg | Dry mass + all four drums' regolith payload. |
| Traversal timing mode | calculated | Calculated route duration. |
| Loaded outbound route | 26.0 s | Includes route turns; excludes deposition and relocation. |
| Empty return route | 26.0 s | Includes route turns; excludes deposition and relocation. |
| Actual route peak speed | 1.00 m/s | Calculated profile, not measured vehicle performance. |
| Score · complete-cycle estimate | 1,465.98 points | Uses complete deliveries and idle remainder; separate from the continuous MDO objective. |
| Empty rover · 30° climb | No climbing solution | Force shortfall per wheel: 31.42 N. Acceptance: solved slip strictly below 1. |

| Prediction | Value | Meaning |
|---|---|---|
| Full rover · 20° climb | No climbing solution | Force shortfall per wheel: 48.65 N.<br>Acceptance: solved slip strictly below 1. |

## Dry mass breakdown

Each assembly is counted once.

| | |
|---|---|
| Equipment allowance | **6.00 kg** |
| Frame | **5.21 kg** |
| Four bare wheels | **5.88 kg** |
| Four drive assemblies | **4.15 kg** |
| Four drums + two drive assemblies | **13.9 kg** |
| Battery | **1.85 kg** |
| Arms + actuators | **6.06 kg** |

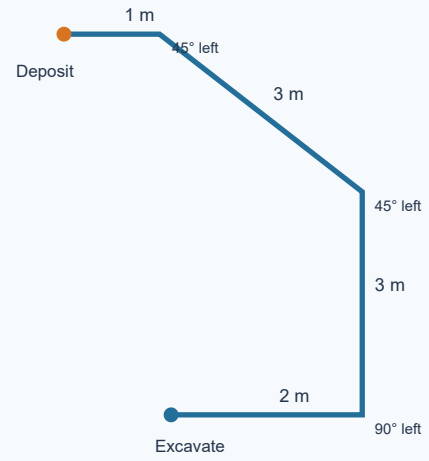
## Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

| | |
|---|---|
| Digging | **8.22 Wh** |
| Dumping | **0.16 Wh** |
| Travel + deposition turn | **2.03 Wh** |
| Excavation actuators | **1.65 Wh** |
| Deposition pose changes | **1.17 Wh** |
| Fresh-site move | **0.07 Wh** |
| Electronics | **2.10 Wh** |

## Route & cycle accounting

25.984 s outbound + 25.984 s return




Return follows the same path in reverse order; yaw signs reverse.

### 25.984 s outbound + 25.984 s return

Lower both pairs together → drive the excavation pass → raise both pairs together → outbound → move both pairs to deposition → dump one pair → turn → dump second pair → return both pairs to travel → return route → relocate to a fresh lane.

Each excavation arm phase lasts as long as the slower arm needs at its actual digging position. Lowering and cutting do not overlap. Maximum extension remains a separate load/capability check. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 302.4 s.

03 / GAPS TO RESOLVE

# Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Ground clearance: lower axle after wheel/bar deductions**<br>`ground_clearance_margin` | Reserve: -2.30 mm | Reserve ≥ 0 | -2.30 mm | Fail |
| **Empty rover: 30° climb**<br>`empty_slope_traversal_margin` | No climbing solution<br>`Slip = 1 is a failure sentinel` | Slip < 1 (strict) | -0.34 | Fail |
| **Full rover: 20° climb**<br>`loaded_slope_traversal_margin` | No climbing solution<br>`Slip = 1 is a failure sentinel` | Slip < 1 (strict) | -0.13 | Fail |
| **Turning: traction reserve**<br>`turn_traction_margin` | Reserve: -160.4 N | Reserve ≥ 0 | -160.4 N | Fail |
| **Travel drum height above wheel-diameter step**<br>`wheel_first_margin` | Reserve: -24.4 mm | Reserve ≥ 0 | -24.4 mm | Fail |
| **Startup position margin**<br>`startup_position_margin` | Reserve: -14.7 mm | Reserve ≥ 0 | -14.7 mm | Fail |
| **Structural stress envelope**<br>`structure_stress_margin` | 148.4 MPa | ≤ 120.0 MPa | -28.4 MPa | Fail |
| **Slope: minimum normal wheel reaction**<br>`slope_wheel_load_margin` | Reserve: -14.2 N | Reserve ≥ 0 | -14.2 N | Fail |
| **Dry rover mass**<br>`dry_mass` | 43.0 kg | ≤ 80.0 kg | 37.0 kg | Pass |
| **Compact overall length**<br>`rover_length` | 1,482 mm | ≤ 1,500 mm | 18.0 mm | Pass |
| **Overall width**<br>`rover_width` | 742.0 mm | ≤ 750.0 mm | 8.00 mm | Pass |
| **Compact overall height**<br>`rover_height` | 738.7 mm | ≤ 750.0 mm | 11.3 mm | Pass |
| **Excavation shaft-power reserve**<br>`drum_power_margin` | Reserve: 1,080 W | Reserve ≥ 0 | 1,080 W | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Excavation: penetration reserve**<br>`penetration_margin` | Reserve: 331.5 N | Reserve ≥ 0 | 331.5 N | Pass |
| **Inlet depth: radial geometry reserve**<br>`cut_depth_margin` | Reserve: 19.9 mm | Reserve ≥ 0 | 19.9 mm | Pass |
| **Ground contact pressure**<br>`ground_pressure_margin` | 20.6 kPa | ≤ 25.0 kPa | 4.44 kPa | Pass |
| **Straight travel: motor-force reserve**<br>`straight_motor_margin` | Reserve: 397.6 N | Reserve ≥ 0 | 397.6 N | Pass |
| **Turning: motor-force reserve**<br>`turn_motor_margin` | Reserve: 5,311 N | Reserve ≥ 0 | 5,311 N | Pass |
| **Straight travel: traction reserve**<br>`straight_traction_margin` | Reserve: 174.5 N | Reserve ≥ 0 | 174.5 N | Pass |
| **Straight travel: shaft-power reserve**<br>`straight_power_margin` | Reserve: 2,208 W | Reserve ≥ 0 | 2,208 W | Pass |
| **Turning: shaft-power reserve**<br>`turn_power_margin` | Reserve: 2,509 W | Reserve ≥ 0 | 2,509 W | Pass |
| **Battery usable energy**<br>`battery_energy_margin` | 45.8 Wh | ≤ 207.4 Wh | 161.5 Wh | Pass |
| **Battery peak-power capacity**<br>`battery_power_margin` | 679.5 W | ≤ 1,728 W | 1,049 W | Pass |
| **Time remaining after one complete cycle**<br>`complete_cycle_margin` | Reserve: 597.6 s | Reserve ≥ 0 | 597.6 s | Pass |
| **Pit depth margin**<br>`pit_depth_margin` | Reserve: 218.4 mm | Reserve ≥ 0 | 218.4 mm | Pass |
| **Transport clearance margin**<br>`transport_clearance_margin` | Reserve: 299.7 mm | Reserve ≥ 0 | 299.7 mm | Pass |
| **Deposition pose: triangle reach reserve**<br>`deposition_reach_margin` | Reserve: 19.7 mm | Reserve ≥ 0 | 19.7 mm | Pass |
| **Deposition command below maximum extension**<br>`deposition_position_margin` | Reserve: 203.2 mm | Reserve ≥ 0 | 203.2 mm | Pass |
| **Deposition drum above ground**<br>`deposition_clearance_margin` | Reserve: 455.1 mm | Reserve ≥ 0 | 455.1 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Dig position margin**<br>`dig_position_margin` | Reserve: 151.0 mm | Reserve ≥ 0 | 151.0 mm | Pass |
| **Pivot spacing margin**<br>`pivot_spacing_margin` | Reserve: 63.2 mm | Reserve ≥ 0 | 63.2 mm | Pass |
| **Pivot mount margin**<br>`pivot_mount_margin` | Reserve: 335.0 mm | Reserve ≥ 0 | 335.0 mm | Pass |
| **Pivot longitudinal margin**<br>`pivot_longitudinal_margin` | Reserve: 165.0 mm | Reserve ≥ 0 | 165.0 mm | Pass |
| **Actuator pivot longitudinal margin**<br>`actuator_pivot_longitudinal_margin` | Reserve: 195.0 mm | Reserve ≥ 0 | 195.0 mm | Pass |
| **Bottom mount margin**<br>`bottom_mount_margin` | Reserve: 52.5 mm | Reserve ≥ 0 | 52.5 mm | Pass |
| **Stroke margin**<br>`stroke_margin` | Reserve: 96.3 mm | Reserve ≥ 0 | 96.3 mm | Pass |
| **Triangle retracted margin**<br>`triangle_retracted_margin` | Reserve: 86.6 mm | Reserve ≥ 0 | 86.6 mm | Pass |
| **Triangle extended margin**<br>`triangle_extended_margin` | Reserve: 33.4 mm | Reserve ≥ 0 | 33.4 mm | Pass |
| **Toggle margin**<br>`toggle_margin` | Reserve: 263.2 mm | Reserve ≥ 0 | 263.2 mm | Pass |
| **Actuator toggle margin**<br>`actuator_toggle_margin` | Reserve: 280.0 mm | Reserve ≥ 0 | 280.0 mm | Pass |
| **Wheel clearance margin**<br>`wheel_clearance_margin` | Reserve: 8.14 mm | Reserve ≥ 0 | 8.14 mm | Pass |
| **Deployed height margin**<br>`deployed_height_margin` | Reserve: 1,606 mm | Reserve ≥ 0 | 1,606 mm | Pass |
| **Actuator force capacity**<br>`actuator_force_margin` | 3,333 N | ≤ 4,003 N | 670.0 N | Pass |
| **Excavation torque reserve**<br>`drum_torque_margin` | Reserve: 221.7 N·m | Reserve ≥ 0 | 221.7 N·m | Pass |
| **Dumping torque reserve**<br>`dump_torque_margin` | Reserve: 85.4 N·m | Reserve ≥ 0 | 85.4 N·m | Pass |
| **Minimum payload reserve**<br>`payload_margin` | Reserve: 123.9 kg | Reserve ≥ 0 | 123.9 kg | Pass |
| **Excavated trench diameter reserve**<br>`trench_traverse_margin` | Reserve: 305.0 mm | Reserve ≥ 0 | 305.0 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Fresh site margin**<br>`fresh_site_margin` | Reserve: 60.0 mm | Reserve ≥ 0 | 60.0 mm | Pass |
| **Frame section margin**<br>`frame_section_margin` | Reserve: 7.25 mm | Reserve ≥ 0 | 7.25 mm | Pass |
| **Arm section margin**<br>`arm_section_margin` | Reserve: 11.1 mm | Reserve ≥ 0 | 11.1 mm | Pass |
| **Structural deflection envelope**<br>`structure_deflection_margin` | 1.63 mm | ≤ 5.00 mm | 3.37 mm | Pass |
| **Structural buckling reserve**<br>`structure_buckling_margin` | Reserve: 5,321 N | Reserve ≥ 0 | 5,321 N | Pass |
| **Level-route dynamic tip reserve**<br>`mobility_stability_margin` | Reserve: 257.5 mm | Reserve ≥ 0 | 257.5 mm | Pass |
| **Asymmetric dump tip reserve**<br>`tip_stability_margin` | Reserve: 21.0 mm | Reserve ≥ 0 | 21.0 mm | Pass |
| **Rock pose: geometric reach reserve**<br>`obstacle_pose_margin` | Reserve: 226.4 mm | Reserve ≥ 0 | 226.4 mm | Pass |
| **Rock endpoint: global tip reserve**<br>`obstacle_tip_margin` | Reserve: 172.3 mm | Reserve ≥ 0 | 172.3 mm | Pass |
| **Obstacle drum–wheel clearance**<br>`obstacle_drum_wheel_margin` | 34.0 mm | ≥ 15.0 mm | 19.0 mm | Pass |
| **30° slope: static tip reserve**<br>`slope_tip_margin` | Reserve: 18.1 mm | Reserve ≥ 0 | 18.1 mm | Pass |
| **Rock path: global tip reserve**<br>`rock_sweep_tip_margin` | Reserve: 172.3 mm | Reserve ≥ 0 | 172.3 mm | Pass |
| **Rock path: smallest wheel reaction**<br>`rock_sweep_wheel_load_margin` | Reserve: 125.3 N | Reserve ≥ 0 | 125.3 N | Pass |
| **Available excavation area**<br>`excavation_area_margin` | Reserve: 0.06 $m^2$ | Reserve ≥ 0 | 0.06 $m^2$ | Pass |
| **Fresh-site move: drive/traction reserve**<br>`site_drive_margin` | Reserve: 167.4 N | Reserve ≥ 0 | 167.4 N | Pass |
| **Plate geometry reserve**<br>`plate_geometry_margin` | Reserve: 22.1 mm | Reserve ≥ 0 | 22.1 mm | Pass |
| **Frame packaging reserve**<br>`frame_packing_margin` | Reserve: 87.7 mm | Reserve ≥ 0 | 87.7 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Plate strip stress**<br>`plate_stress_margin` | 29.7 MPa | ≤ 120.0 MPa | 90.3 MPa | Pass |
| **Plate strip buckling reserve**<br>`plate_buckling_margin` | Reserve: 5,321 N | Reserve ≥ 0 | 5,321 N | Pass |
| **aluminum cross-tube bending stress**<br>`cross_tube_bending_margin` | 72.4 MPa | ≤ 120.0 MPa | 47.6 MPa | Pass |
| **Static wheel equilibrium and regular-branch reserve**<br>`wheel_equilibrium_margin` | Reserve: 0.82 | Reserve ≥ 0 | 0.82 | Pass |
| **Actuator catalog stroke domain**<br>`actuator_stroke_domain_margin` | Reserve: 152.4 mm | Reserve ≥ 0 | 152.4 mm | Pass |
| **Actuator load-curve domain**<br>`actuator_curve_margin` | Reserve: 670.0 N | Reserve ≥ 0 | 670.0 N | Pass |
| **Forward advance per scoop: inlet-depth reserve**<br>`feed_margin` | 15.0 mm | ≤ 25.0 mm | 10.0 mm | Pass |
| **Moving excavation: operating-pose reach**<br>`operating_reach_margin` | Reserve: 39.0 mm | Reserve ≥ 0 | 39.0 mm | Pass |
| **Plunge/recovery: axle reaches ground level**<br>`axle_ground_margin` | -52.1 mm | ≤ 0.00 mm | 52.1 mm | Pass |
| **Moving excavation: geometry domain**<br>`excavation_domain_margin` | Reserve: 154.5 mm | Reserve ≥ 0 | 154.5 mm | Pass |
| **Excavation feed: motor-force reserve**<br>`feed_motor_margin` | Reserve: 7,129 N | Reserve ≥ 0 | 7,129 N | Pass |
| **Excavation feed: shaft-power reserve**<br>`feed_power_margin` | Reserve: 2,669 W | Reserve ≥ 0 | 2,669 W | Pass |
| **Lin act side arm offset search range**<br>`lin_act_side_arm_offset_bounds` | -50.8 mm | -500.0 mm to -50.8 mm | 0.00 mm | Pass |
| **Lin act top arm offset search range**<br>`lin_act_top_arm_offset_bounds` | 60.0 mm | -150.0 mm to 150.0 mm | 90.0 mm | Pass |
| **Wheel diameter search range**<br>`wheel_diameter_bounds` | 355.0 mm | 180.0 mm to 359.0 mm | 4.00 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Wheel width search range**<br>`wheel_width_bounds` | 152.0 mm | 50.0 mm to 359.0 mm | 102.0 mm | Pass |
| **Drum outer diameter search range**<br>`drum_diameter_bounds` | 359.0 mm | 160.0 mm to 359.0 mm | 0.00 mm | Pass |
| **Width of each drum search range**<br>`drum_length_bounds` | 340.0 mm | 120.0 mm to 359.0 mm | 19.0 mm | Pass |
| **Excavation drum RPM search range**<br>`drum_rpm_bounds` | 30.0 rpm | 5.00 rpm to 120.0 rpm | 25.0 rpm | Pass |
| **Straight excavation feed speed search range**<br>`feed_speed_bounds` | 0.03 m/s | 0.0001 m/s to 0.50 m/s | 0.03 m/s | Pass |
| **Dumping speed search range**<br>`dump_rpm_bounds` | 60.0 rpm | 5.00 rpm to 60.0 rpm | 0.00 rpm | Pass |
| **Drum reduction ratio search range**<br>`drum_gear_ratio_bounds` | 60.0 | 9.00 to 125.0 | 51.0 | Pass |
| **Inlet depth / fresh-layer cut depth search range**<br>`inlet_depth_bounds` | 25.0 mm | 3.00 mm to 25.0 mm | 0.00 mm | Pass |
| **Arm wall thickness search range**<br>`arm_wall_bounds` | 1.59 mm | 1.00 mm to 5.00 mm | 0.59 mm | Pass |
| **Battery capacity search range**<br>`battery_capacity_bounds` | 259.2 Wh | 10.0 Wh to 300.0 Wh | 40.8 Wh | Pass |
| **Travel actuator length search range**<br>`travel_actuator_length_bounds` | 470.0 mm | 403.1 mm to 606.3 mm | 66.9 mm | Pass |
| **Deposition actuator length search range**<br>`deposition_actuator_length_bounds` | 403.1 mm | 403.1 mm to 606.3 mm | 0.00 mm | Pass |
| **Compact actuator length search range**<br>`actuator_retracted_length_bounds` | 455.3 mm | 403.1 mm to 606.3 mm | 52.2 mm | Pass |
| **Maximum-extension actuator length search range**<br>`actuator_extended_length_bounds` | 606.3 mm | 403.1 mm to 606.3 mm | 0.00 mm | Pass |
| **Actuator pivot height search range**<br>`top_pivot_height_bounds` | 370.0 mm | 350.0 mm to 700.0 mm | 20.0 mm | Pass |
| **Actuator pivot longitudinal position search range**<br>`actuator_pivot_x_bounds` | 80.0 mm | 20.0 mm to 300.0 mm | 60.0 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Arm pivot height search range**<br>`bottom_pivot_height_bounds` | 230.0 mm | 90.0 mm to 300.0 mm | 70.0 mm | Pass |
| **Arm length search range**<br>`arm_length_bounds` | 558.8 mm | 300.0 mm to 700.0 mm | 141.2 mm | Pass |
| **Arm pivot longitudinal position search range**<br>`pivot_x_bounds` | 110.0 mm | 50.0 mm to 400.0 mm | 60.0 mm | Pass |
| **Wheelbase search range**<br>`wheelbase_bounds` | 550.0 mm | 380.0 mm to 650.0 mm | 100.0 mm | Pass |
| **Wheel-center track search range**<br>`track_width_bounds` | 590.0 mm | 500.0 mm to 699.0 mm | 90.0 mm | Pass |
| **Aluminum arm box width search range**<br>`arm_box_width_bounds` | 50.8 mm | 25.0 mm to 100.0 mm | 25.8 mm | Pass |
| **Aluminum arm box height search range**<br>`arm_box_height_bounds` | 50.8 mm | 25.0 mm to 100.0 mm | 25.8 mm | Pass |
| **aluminum cross-tube span / plate spacing search range**<br>`cross_tube_span_bounds` | 160.0 mm | 160.0 mm to 460.0 mm | 0.00 mm | Pass |
| **Aluminum cross-tube wall thickness search range**<br>`cross_tube_wall_bounds` | 3.12 mm | 1.00 mm to 6.00 mm | 2.12 mm | Pass |
| **Plate strip width search range**<br>`plate_spoke_width_bounds` | 18.8 mm | 12.0 mm to 70.0 mm | 6.75 mm | Pass |
| **Rocker box width search range**<br>`rocker_box_width_bounds` | 50.8 mm | 20.0 mm to 65.0 mm | 14.2 mm | Pass |
| **Rocker box height search range**<br>`rocker_box_height_bounds` | 50.8 mm | 25.0 mm to 90.0 mm | 25.8 mm | Pass |
| **Rocker wall search range**<br>`rocker_wall_bounds` | 1.59 mm | 1.00 mm to 5.00 mm | 0.59 mm | Pass |
| **Axle outer diameter search range**<br>`axle_outer_diameter_bounds` | 33.0 mm | 25.0 mm to 90.0 mm | 8.00 mm | Pass |
| **Axle wall search range**<br>`axle_wall_bounds` | 1.00 mm | 1.00 mm to 6.00 mm | 0.00 mm | Pass |
| **Rocker pivot above wheel centers search range**<br>`rocker_pivot_offset_bounds` | 96.2 mm | 20.0 mm to 150.0 mm | 53.8 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Drum-supported recovery driving** | Axle-to-ground reach is screened geometrically. Drum traction, motor-ground clearance, recovery loads and actual escape capability require hardware and soil tests. | — | — | Not assessed |
| **Slope motor limits / sideways sliding** | The separate climbing checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated. | — | — | Not assessed |
| **Real 30 cm rock traversal** | Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence. | — | — | Not assessed |
| **50 cm ditch crossing** | Ditch width/radius and contact sequence have not been defined. | — | — | Not assessed |
| **Continuous motor operation** | Stator-current limits and thermal duty remain undecided. | — | — | Not assessed |
| **Connections & material validation** | Bonded inserts, bushings, bolts, local failure and selected aluminum cross-tube properties need separate checks. | — | — | Not assessed |
| **Soil and skid-steer predictions** | Rolling, traction, capture and scrub relationships remain uncalibrated. | — | — | Not assessed |

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

# Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

| Quantity / scope | Prediction | Load case / interpretation |
|---|---|---|
| Required force per actuator<br>`required_actuator_force` | 3,333 N | Quasi-static gravity envelope for loaded drums and arm; extraction resistance and dynamic loads are not modeled. |
| Frame gravity reaction at each actuator<br>`frame_gravity_actuator_force` | 5,000 N | Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load. |
| Nominal force per wheel<br>`frame_nominal_wheel_reaction` | 409.3 N | Equal-share level-ground weight. Not a rock or slope wheel-load envelope. |
| Total cutting force<br>`cut_force` | 181.2 N | Across four drums at the selected bite and engagement. |
| Worst modeled contact pressure<br>`ground_pressure` | 20.6 kPa | Route pressure estimate, including load transfer and rocker factor. |
| Structural stress envelope<br>`structure_stress` | 148.4 MPa | Envelope of local member stresses; not one assembled-frame FEA result. |
| Structural deflection envelope<br>`structure_deflection` | 1.63 mm | Conservative combination of local/relative deflections. |
| Upper/lower mount relative displacement<br>`frame_mount_relative_deflection` | 0.32 mm | Frame gravity sweep. |
| Rocker-end flexure<br>`rocker_deflection` | 0.46 mm | Local beam-screen displacement. |
| Suspension-axle flexure<br>`axle_deflection` | 0.44 mm | Axle-overhang screen. |
| Arm-tip flexure<br>`arm_deflection` | 1.63 mm | Relative arm beam displacement. |
| Loaded CG height<br>`loaded_cg_z` | 429.0 mm | Fully loaded travel pose, above level ground. |

APPENDIX / NATIVE UNITS

# Design inputs & output data

154 inputs supplied by file; 1 CLI overrides; 0 remaining inputs use model defaults. Legacy breakout_force is inactive; actuator loads use gravity only. Values below retain their native model units.

## All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `7e376f97af55b6f4e550d59b0dabdd15f9736d35e69ca89fce75fe33b0fd5e7b`.

| Input port | Effective value | Native units |
|---|---|---|
| lin_act_side_arm_offset | -0.0508 | m |
| lin_act_top_arm_offset | 0.06 | m |
| wheel_diameter | 0.355 | m |
| wheel_width | 0.152 | m |
| drum_diameter | 0.359 | m |
| drum_length | 0.34 | m |
| drum_rpm | 30 | rpm |
| feed_speed | 0.03 | m/s |
| dump_rpm | 60 | rpm |
| drum_gear_ratio | 60 | — |
| inlet_depth | 0.025 | m |
| arm_wall | 0.0015875 | m |
| battery_capacity | 259.2 | W*h |
| travel_actuator_length | 0.47 | m |
| deposition_actuator_length | 0.403098 | m |
| actuator_retracted_length | 0.455298 | m |
| actuator_extended_length | 0.606298 | m |
| top_pivot_height | 0.37 | m |
| actuator_pivot_x | 0.08 | m |
| bottom_pivot_height | 0.23 | m |
| arm_length | 0.5588 | m |
| pivot_x | 0.11 | m |
| wheelbase | 0.55 | m |
| track_width | 0.59 | m |
| arm_box_width | 0.0508 | m |
| arm_box_height | 0.0508 | m |
| cross_tube_span | 0.16 | m |
| cross_tube_wall | 0.003125 | m |
| plate_spoke_width | 0.01875 | m |
| rocker_box_width | 0.0508 | m |
| rocker_box_height | 0.0508 | m |
| rocker_wall | 0.0015875 | m |
| axle_outer_diameter | 0.033 | m |

| Input port | Effective value | Native units |
|---|---|---|
| axle_wall | 0.001 | m |
| rocker_pivot_offset | 0.0962 | m |
| minimum_ground_clearance | 0.1524 | m |
| clearance_radius_deflection_fraction | 0.2 | — |
| grouser_height | 0.018 | m |
| tipping_radius_deflection_fraction | 0.2 | — |
| tipping_reference_fill_fraction | 0.9 | — |
| wheel_compression_hard_stop | 0.064 | m |
| differential_bar_clearance_allowance | 0.0254 | m |
| empty_slope_degrees | 30 | deg |
| loaded_slope_degrees | 20 | deg |
| gravity | 9.80665 | m/s**2 |
| run_time | 900 | s |
| traversal_time_one_way | 45 | s |
| travel_speed | 1 | m/s |
| acceleration | 0.4 | m/s**2 |
| turn_angle | 3.1415927 | rad |
| overhead_time | 8 | s |
| electronics_power | 25 | W |
| actuator_idle_power | 2 | W |
| soil_density | 1500 | kg/m**3 |
| berm_density | 1500 | kg/m**3 |
| fill_fraction | 0.9 | — |
| delivery_efficiency | 0.9 | — |
| capture_fraction | 0.65 | — |
| cut_pressure | 18000 | Pa |
| internal_lift_factor | 0.15 | — |
| vertical_cut_fraction | 0.5 | — |
| force_imbalance | 0.15 | — |
| dump_revolutions | 3 | — |
| drum_efficiency | 0.7 | — |
| drum_gear_efficiency | 0.8 | — |
| drive_efficiency | 0.72 | — |
| drive_gear_efficiency | 0.8 | — |
| motor_rated_rpm | 3000 | rpm |
| traction_coefficient | 0.6 | — |
| rocker_load_factor | 1.2 | — |
| wheel_deflection_fraction | 0.07 | — |
| rolling_base | 0.04 | — |
| rolling_pressure_scale | 200000 | Pa |
| allowable_ground_pressure | 25000 | Pa |
| scrub_coefficient | 0.25 | — |
| turn_speed_fraction | 0.35 | — |
| turn_accel_fraction | 0.5 | — |
| chassis_height | 0.35 | m |
| drum_pair_gap | 0.06 | m |

| Input port | Effective value | Native units |
|---|---|---|
| fixed_mass | 6 | kg |
| wheel_areal_mass | 4 | kg/m**2 |
| drum_areal_mass | 5 | kg/m**2 |
| motor_mass_per_watt | 0.006 | kg/W |
| gear_mass_per_torque | 0.05 | kg/(N*m) |
| drive_fixed_mass | 0.35 | kg |
| drive_motor_cg_offset | 0.05842 | m |
| drum_fixed_mass | 0.5 | kg |
| battery_specific_energy | 140 | W*h/kg |
| battery_usable_fraction | 0.8 | — |
| battery_c_rate | 6.667 | 1/h |
| mass_score_weight | 1 | — |
| energy_score_weight | 1 | — |
| transport_clearance | 0.08 | m |
| mechanism_clearance | 0.015 | m |
| minimum_pivot_spacing | 0.08 | m |
| minimum_stroke | 0.04 | m |
| actuator_housing_allowance | 0.04 | m |
| pit_depth | 0.45 | m |
| actuator_efficiency | 0.65 | — |
| actuator_safety_factor | 1 | — |
| breakout_force | 100 | N |
| actuator_fixed_mass | 0.6 | kg |
| actuator_mass_per_force | 0.0004 | kg/N |
| actuator_mass_per_power | 0.003 | kg/W |
| actuator_mass_per_length | 1 | kg/m |
| stored_soil_density | 1500 | kg/m**3 |
| usable_volume_fraction | 0.9 | — |
| scoops_per_revolution | 4 | — |
| max_dig_froude | 0.3 | — |
| max_dump_froude | 0.2 | — |
| drum_spinup_time | 2 | s |
| drum_torque_service_factor | 1.25 | — |
| minimum_dig_depth | 0.005 | m |
| minimum_payload | 0.05 | kg |
| fresh_site_distance | 0.8 | m |
| site_speed | 0.1 | m/s |
| available_excavation_area | 12 | m**2 |
| tube_density | 2700 | kg/m**3 |
| tube_youngs_modulus | 6.9e+10 | Pa |
| tube_yield_strength | 2.4e+08 | Pa |
| structure_safety_factor | 2 | — |
| structure_load_factor | 1.5 | — |
| allowable_deflection | 0.005 | m |
| joint_mass | 2 | kg |
| differential_mass | 1 | kg |

| Input port | Effective value | Native units |
|---|---|---|
| obstacle_height | 0.3 | m |
| slope_angle | 0.52359878 | rad |
| frame_tube_diameter | 0.038 | m |
| frame_wall | 0.003175 | m |
| arm_tube_diameter | 0.05 | m |
| plate_thickness | 0.00635 | m |
| cross_tube_inner_diameter | 0.03175 | m |
| wheel_rocker_gap | 0.0127 | m |
| plate_hub_radius | 0.03 | m |
| plate_cutout_clearance | 0.005 | m |
| frame_mount_clearance | 0.003 | m |
| minimum_axle_overhang | 0.01 | m |
| plate_buckling_k | 1 | — |
| cross_tube_density | 2700 | kg/m**3 |
| cross_tube_youngs_modulus | 6.9e+10 | Pa |
| cross_tube_yield_strength | 2.4e+08 | Pa |
| drive_motor_power | 835.4 | W |
| drum_motor_power | 835.4 | W |
| drive_gear_ratio | 100 | — |
| motor_bus_voltage | 12 | V |
| x44_motor_mass | 0.34019428 | kg |
| drive_gearbox_mass | 0.697 | kg |
| drum_gearbox_mass | 0.753 | kg |
| drive_stator_current_limit | 0 | A |
| drum_stator_current_limit | 0 | A |
| actuator_force_rating | 4003.3995 | N |
| actuator_speed | 0.009906 | m/s |
| actuator_catalog_stroke | 0.2032 | m |
| actuator_installation_allowance | 0 | m |

## Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

| Port | Lower | Evaluated | Upper | Native units |
|---|---|---|---|---|
| lin_act_side_arm_offset | -0.5 | -0.0508 | -0.0508 | m |
| lin_act_top_arm_offset | -0.15 | 0.06 | 0.15 | m |
| wheel_diameter | 0.18 | 0.355 | 0.359 | m |
| wheel_width | 0.05 | 0.152 | 0.359 | m |
| drum_diameter | 0.16 | 0.359 | 0.359 | m |
| drum_length | 0.12 | 0.34 | 0.359 | m |
| drum_rpm | 5 | 30 | 120 | rpm |
| feed_speed | 0.0001 | 0.03 | 0.5 | m/s |
| dump_rpm | 5 | 60 | 60 | rpm |
| drum_gear_ratio | 9 | 60 | 125 | — |
| inlet_depth | 0.003 | 0.025 | 0.025 | m |
| arm_wall | 0.001 | 0.0015875 | 0.005 | m |
| battery_capacity | 10 | 259.2 | 300 | W*h |
| travel_actuator_length | 0.403098 | 0.47 | 0.606298 | m |
| deposition_actuator_length | 0.403098 | 0.403098 | 0.606298 | m |
| actuator_retracted_length | 0.403098 | 0.455298 | 0.606298 | m |
| actuator_extended_length | 0.403098 | 0.606298 | 0.606298 | m |
| top_pivot_height | 0.35 | 0.37 | 0.7 | m |
| actuator_pivot_x | 0.02 | 0.08 | 0.3 | m |
| bottom_pivot_height | 0.09 | 0.23 | 0.3 | m |
| arm_length | 0.3 | 0.5588 | 0.7 | m |
| pivot_x | 0.05 | 0.11 | 0.4 | m |
| wheelbase | 0.38 | 0.55 | 0.65 | m |
| track_width | 0.5 | 0.59 | 0.699 | m |
| arm_box_width | 0.025 | 0.0508 | 0.1 | m |
| arm_box_height | 0.025 | 0.0508 | 0.1 | m |
| cross_tube_span | 0.16 | 0.16 | 0.46 | m |
| cross_tube_wall | 0.001 | 0.003125 | 0.006 | m |
| plate_spoke_width | 0.012 | 0.01875 | 0.07 | m |
| rocker_box_width | 0.02 | 0.0508 | 0.065 | m |
| rocker_box_height | 0.025 | 0.0508 | 0.09 | m |
| rocker_wall | 0.001 | 0.0015875 | 0.005 | m |
| axle_outer_diameter | 0.025 | 0.033 | 0.09 | m |
| axle_wall | 0.001 | 0.001 | 0.006 | m |
| rocker_pivot_offset | 0.02 | 0.0962 | 0.15 | m |

## All output values

| Output quantity | Value | Native units |
|---|---|---|
| dry_mass | 43.048775 | kg |
| payload_mass | 123.91447 | kg |
| wheel_mass | 5.8796791 | kg |
| drum_mass | 13.904561 | kg |
| drive_mass | 4.1487771 | kg |
| battery_mass | 1.8514286 | kg |
| rover_length | 1.482048 | m |
| rover_width | 0.742 | m |
| rover_height | 0.73871652 | m |
| ground_pressure | 20563.209 | Pa |
| actual_travel_speed | 1 | m/s |
| dig_time | 176.93691 | s |
| travel_time | 51.967957 | s |
| dump_time | 10 | s |
| turn_time | 4.3979138 | s |
| cycle_time | 302.42597 | s |
| cycle_energy | 15.396447 | W*h |
| cycles | 2.9759349 | — |
| berm_volume | 0.22125684 | m**3 |
| run_energy | 45.818823 | W*h |
| average_power | 183.27529 | W |
| peak_power | 679.46031 | W |
| mass_points | 1507.6388 | — |
| energy_points | 482.89506 | — |
| construction_points | 1990.5339 | — |
| no_load_speed | 1.4419647 | m/s |
| loaded_motor_speed_limit | 1.0210198 | m/s |
| loaded_power_speed_limit | 5.5336824 | m/s |
| stall_drive_force | 7415.6297 | N |
| actuation_time | 19.322293 | s |
| lowering_time | 13.759136 | s |
| arm_energy | 5932.03 | J |
| arm_peak_power | 336 | W |
| deposition_reach_margin | 0.019745175 | m |
| deposition_position_margin | 0.2032 | m |
| deposition_clearance_margin | 0.45514289 | m |
| required_actuator_force | 3333.3548 | N |
| linkage_mass | 6.0564014 | kg |
| stroke | 0.136298 | m |
| stowed_x | 0.561524 | m |
| stowed_y | 0.55921652 | m |
| travel_x | 0.59353307 | m |
| travel_y | 0.51009501 | m |

| Output quantity | Value | Native units |
|---|---|---|
| deposition_x | 0.38817028 | m |
| deposition_y | 0.71464289 | m |
| extended_x | 0.5923666 | m |
| extended_y | -0.052099104 | m |
| sweep_x_min | 0.38817028 | m |
| sweep_x_max | 0.6688 | m |
| actuator_leverage_max | 4.9158482 | — |
| actuator_attachment_distance | 0.51153104 | m |
| arm_physical_length | 0.5588 | m |
| arm_extension_mass | 0 | kg |
| actuator_body_margin | 0 | m |
| chassis_clearance_margin | -0.086434025 | m |
| stowed_length | 1.482048 | m |
| stowed_height | 0.73871652 | m |
| deployed_length | 1.6965923 | m |
| deployed_height | 0.89414289 | m |
| route_distance | 18 | m |
| route_turn_angle | 360 | deg |
| route_translation_time | 37.268827 | s |
| route_turn_time | 14.69913 | s |
| route_peak_acceleration | 0.4 | m/s**2 |
| deposition_actuation_time | 13.283197 | s |
| deposition_arm_energy | 4200.8733 | J |
| deposition_arm_peak_power | 336 | W |
| payload_volume | 0.082609647 | m**3 |
| fresh_volume | 0.12709176 | m**3 |
| storage_capacity | 0.082609647 | m**3 |
| reachable_soil_mass | 123.91447 | kg |
| target_payload_mass | 123.91447 | kg |
| trench_depth | 0.05 | m |
| excavation_footprint | 4.0110645 | m**2 |
| cut_force | 181.24161 | N |
| drum_froude | 0.18065231 | — |
| dig_overlap_time | 0 | s |
| dig_power | 310.3608 | W |
| dump_power | 148.84199 | W |
| dig_energy | 29576.754 | J |
| dump_energy | 585.74483 | J |
| drum_power_margin | 1080.3915 | W |
| drum_torque_margin | 221.69291 | N*m |
| dump_torque_margin | 85.367155 | N*m |
| dig_froude_margin | 0.11934769 | — |
| dump_froude_margin | -0.52260925 | — |
| penetration_margin | 331.54346 | N |
| dig_traction_margin | 121.01926 | N |
| cut_depth_margin | 0.019875 | m |

| Output quantity | Value | Native units |
|---|---|---|
| dig_reach_margin | 0.03897956 | m |
| payload_margin | 123.86447 | kg |
| structure_mass | 5.2079282 | kg |
| rocker_mass | 0.82096711 | kg |
| arm_linear_mass | 0.84374831 | kg/m |
| arm_envelope_radius | 0.035921024 | m |
| frame_section_margin | 0.00725 | m |
| arm_section_margin | 0.0111125 | m |
| structure_stress | 1.4843749e+08 | Pa |
| structure_stress_margin | -28437491 | Pa |
| structure_deflection | 0.0016341695 | m |
| structure_deflection_margin | 0.0033658305 | m |
| structure_buckling_margin | 5321.3286 | N |
| empty_yaw_inertia | 8.6234809 | kg*m**2 |
| empty_cg_x | 1.5473932e-17 | m |
| empty_cg_z | 0.36786246 | m |
| loaded_yaw_inertia | 59.424684 | kg*m**2 |
| loaded_cg_x | 0 | m |
| loaded_cg_z | 0.42901637 | m |
| half_yaw_inertia | 34.024082 | kg*m**2 |
| half_cg_x | 0.35020536 | m |
| half_cg_z | 0.41648089 | m |
| loaded_sprung_mass | 156.11382 | kg |
| loaded_sprung_cg_x | 0 | m |
| loaded_sprung_cg_z | 0.44443755 | m |
| deposition_empty_yaw_inertia | 5.4019082 | kg*m**2 |
| deposition_empty_cg_x | -1.0315955e-17 | m |
| deposition_empty_cg_z | 0.44688954 | m |
| deposition_loaded_yaw_inertia | 31.221293 | kg*m**2 |
| deposition_loaded_cg_x | 0 | m |
| deposition_loaded_cg_z | 0.60120074 | m |
| deposition_half_yaw_inertia | 18.311601 | kg*m**2 |
| deposition_half_cg_x | 0.2290341 | m |
| deposition_half_cg_z | 0.56956966 | m |
| relocation_time | 18.517698 | s |
| relocation_energy | 253.63583 | J |
| relocation_peak_power | 19.930256 | W |
| fresh_site_margin | 0.06 | m |
| site_drive_margin | 167.40072 | N |
| site_turn_time | 10.267698 | s |
| mobility_energy | 7317.5221 | J |
| drive_peak_power | 654.46031 | W |
| turn_peak_power | 235.85642 | W |
| ground_pressure_margin | 4436.7911 | Pa |
| straight_motor_margin | 397.60089 | N |
| turn_motor_margin | 5311.1354 | N |

| Output quantity | Value | Native units |
|---|---|---|
| straight_traction_margin | 174.544 | N |
| turn_traction_margin | -160.44931 | N |
| straight_power_margin | 2207.8176 | W |
| turn_power_margin | 2509.2124 | W |
| mobility_stability_margin | 0.257501 | m |
| trench_traverse_margin | 0.305 | m |
| tip_stability_margin | 0.045965896 | m |
| tip_worst_mask | 3 | — |
| obstacle_pose_margin | 0.22636042 | m |
| obstacle_tip_margin | 0.17232893 | m |
| obstacle_pose_valid | 1 | — |
| obstacle_evaluated_height | 0.3 | m |
| obstacle_worst_wheel | 0 | — |
| obstacle_chassis_pitch | 0.35089115 | rad |
| obstacle_chassis_roll | 0.2511198 | rad |
| obstacle_loaded_cg_x | -0.053464411 | m |
| obstacle_loaded_cg_y | -0.036294813 | m |
| obstacle_loaded_cg_z | 0.44146845 | m |
| obstacle_drum_wheel_gap_bound | 0.033996069 | m |
| obstacle_drum_wheel_margin | 0.018996069 | m |
| obstacle_clearance_pose_valid | 1 | — |
| obstacle_clearance_evaluated_height | 0.3 | m |
| slope_pitch_tip_margin | 0.018089277 | m |
| slope_roll_tip_margin | 0.041992591 | m |
| slope_tip_margin | 0.018089277 | m |
| slope_tip_angle | 0.55389571 | rad |
| rock_sweep_tip_margin | 0.17232893 | m |
| rock_sweep_worst_height | 0.3 | m |
| rock_sweep_raised_wheel | 0 | — |
| rock_sweep_pitch | 0.35089115 | rad |
| rock_sweep_roll | 0.2511198 | rad |
| rock_sweep_cg_x | -0.053464411 | m |
| rock_sweep_cg_y | -0.036294813 | m |
| rock_sweep_cg_z | 0.44146845 | m |
| rock_sweep_wheel_load_margin | 125.34833 | N |
| rock_sweep_load_height | 0.3 | m |
| rock_sweep_load_raised_wheel | 0 | — |
| rock_sweep_unloaded_wheel | 0 | — |
| rock_sweep_pose_valid | 1 | — |
| rock_sweep_evaluated_height | 0.3 | m |
| rock_sweep_samples | 41 | — |
| rock_sweep_force_balance_error | 3.2684966e-13 | N |
| empty_slope_slip | 1 | — |
| empty_slope_solution_valid | 0 | — |
| empty_slope_traversal_margin | -0.34375096 | — |
| empty_slope_force_shortfall | 31.419266 | N |

| Output quantity | Value | Native units |
|---|---|---|
| empty_slope_sinkage | 0.0064195488 | m |
| empty_slope_transfer_flag_count | 0 | — |
| loaded_slope_slip | 1 | — |
| loaded_slope_solution_valid | 0 | — |
| loaded_slope_traversal_margin | -0.12647032 | — |
| loaded_slope_force_shortfall | 48.646991 | N |
| loaded_slope_sinkage | 0.016925441 | m |
| loaded_slope_transfer_flag_count | 3 | — |
| bottom_pivot_ground_clearance | 0.1501 | m |
| rocker_pivot_ground_clearance | 0.1963 | m |
| ground_clearance | 0.1501 | m |
| ground_clearance_margin | -0.0023 | m |
| bottom_pivot_axle_radius | 0.019 | m |
| rocker_pivot_axle_radius | 0.0165 | m |
| tipping_reference_mass | 166.96325 | kg |
| tipping_wheel_outer_radius | 0.1775 | m |
| tipping_wheel_tread_radius | 0.1595 | m |
| tipping_wheel_deflection | 0.0355 | m |
| tipping_wheel_deflection_ratio | 0.2 | — |
| tipping_wheel_support_height | 0.142 | m |
| tipping_loaded_cg_z | 0.39351637 | m |
| wheel_spring_compliance | 8.6725495e-05 | m/N |
| slope_wheel_load_margin | -14.216685 | N |
| wheel_equilibrium_margin | 0.81523425 | — |
| slope_worst_heading | 0 | rad |
| slope_load_worst_heading | 0.63130953 | rad |
| slope_contact_limit_angle | 0.50049336 | rad |
| slope_equilibrium_valid | 0 | — |
| slope_pose_error | 2.5337371e-13 | m |
| rock_equilibrium_valid | 1 | — |
| rock_pose_error | 7.4593109e-15 | m |
| slope_fl_normal_load | 870.98827 | N |
| slope_fl_compression | 0.064 | m |
| slope_fl_stop_reaction | 133.02766 | N |
| slope_fr_normal_load | 368.69672 | N |
| slope_fr_compression | 0.031975406 | m |
| slope_fr_stop_reaction | 0 | N |
| slope_rl_normal_load | 192.51849 | N |
| slope_rl_compression | 0.016696261 | m |
| slope_rl_stop_reaction | 0 | N |
| slope_rr_normal_load | -14.216685 | N |
| slope_rr_compression | 0 | m |
| slope_rr_stop_reaction | 0 | N |
| obstacle_equilibrium_valid | 1 | — |
| excavation_lowering_time | 9.8241914 | s |
| excavation_raising_time | 9.4981017 | s |

| Output quantity | Value | Native units |
|---|---|---|
| front_dig_x | 0.66367607 | m |
| feed_per_scoop | 0.015 | m |
| cut_depth | 0.025 | m |
| rear_dig_actuator_length | 0.56731844 | m |
| feed_drive_distance | 5.0613574 | m |
| front_dig_depth | 0.025 | m |
| axle_ground_margin | 0.052099104 | m |
| operating_reach_margin | 0.03897956 | m |
| rear_dig_y | 0.1295 | m |
| excavation_domain_margin | 0.1545 | m |
| rear_dig_depth | 0.05 | m |
| feed_margin | 0.01 | m |
| front_dig_y | 0.1545 | m |
| feed_distance | 3.7379931 | m |
| inlet_width | 0.34 | m |
| front_dig_actuator_length | 0.56162545 | m |
| rear_dig_x | -0.65968827 | m |
| feed_speed_limit | 0.05 | m/s |
| feed_time | 168.93691 | s |
| pair_cutting_time | 124.74977 | s |
| soil_supply_rate | 0.00102 | m**3/s |
| scoop_throughput | 0.0017 | m**3/s |
| feed_drive_energy | 1641.3622 | J |
| feed_drive_peak_power | 13.511638 | W |
| feed_motor_margin | 7129.0686 | N |
| feed_power_margin | 2669.3007 | W |
| nasa_force_source_valid | 0 | — |
| nasa_tip_feed_ratio | 18.797196 | — |
| outbound_travel_time | 25.983979 | s |
| return_travel_time | 25.983979 | s |
| outbound_route_acceleration | 0.4 | m/s**2 |
| return_route_acceleration | 0.4 | m/s**2 |
| outbound_straight_speed_limit | 1 | m/s |
| return_straight_speed_limit | 1 | m/s |
| outbound_slip | 0.22764922 | — |
| return_slip | 0.25584995 | — |
| outbound_sinkage | 0.015032553 | m |
| return_sinkage | 0.005968334 | m |
| outbound_transfer_flag_count | 3 | — |
| return_transfer_flag_count | 0 | — |
| pit_depth_margin | 0.2184009 | m |
| transport_clearance_margin | 0.29971652 | m |
| wheel_first_margin | -0.024404995 | m |
| startup_position_margin | -0.014702 | m |
| dig_position_margin | 0.151 | m |
| pivot_spacing_margin | 0.063178211 | m |

| Output quantity | Value | Native units |
|---|---|---|
| pivot_mount_margin | 0.335 | m |
| pivot_longitudinal_margin | 0.165 | m |
| bottom_mount_margin | 0.0525 | m |
| stroke_margin | 0.096298 | m |
| triangle_retracted_margin | 0.086647175 | m |
| triangle_extended_margin | 0.033411246 | m |
| toggle_margin | 0.26317028 | m |
| wheel_clearance_margin | 0.0081367072 | m |
| deployed_height_margin | 1.6058571 | m |
| actuator_pivot_longitudinal_margin | 0.195 | m |
| actuator_toggle_margin | 0.27995974 | m |
| actuator_catalog_retracted_length | 0.403098 | m |
| actuator_catalog_extended_length | 0.606298 | m |
| actuator_unit_mass | 2.5567141 | kg |
| actuator_stroke_domain_margin | 0.1524 | m |
| actuator_force_margin | 670.04465 | N |
| actuator_curve_margin | 670.04465 | N |
| actuator_curve_valid | 1 | — |
| actuator_loaded_speed | 0.010246094 | m/s |
| actuator_raising_current | 12.326311 | A |
| actuator_lowering_current | 14 | A |
| actuator_raising_time | 13.302436 | s |
| rocker_length | 0.4865 | m |
| plate_mass_total | 0.69711602 | kg |
| cross_tube_mass_total | 0.59164044 | kg |
| suspension_axle_mass | 0.098204678 | kg |
| suspension_bearing_span | 0.3618 | m |
| cross_tube_outer_diameter | 0.038 | m |
| plate_geometry_margin | 0.022055597 | m |
| frame_packing_margin | 0.087725 | m |
| plate_geometry_valid | 1 | — |
| structure_cg_z | 0.28036953 | m |
| structure_sprung_cg_z | 0.28161766 | m |
| structure_yaw_inertia | 0.058394085 | kg*m**2 |
| plate_stress | 29712149 | Pa |
| plate_stress_margin | 90287851 | Pa |
| plate_buckling_margin | 5321.3286 | N |
| cross_tube_bending_stress | 72429327 | Pa |
| cross_tube_bending_margin | 47570673 | Pa |
| frame_mount_relative_deflection | 0.0003181465 | m |
| rocker_deflection | 0.00045775775 | m |
| axle_deflection | 0.00044237718 | m |
| arm_deflection | 0.0016341554 | m |
| rocker_buckling_margin | 181657.61 | N |
| arm_buckling_margin | 29949.062 | N |
| frame_gravity_actuator_force | 5000.0322 | N |

| Output quantity | Value | Native units |
|---|---|---|
| frame_nominal_wheel_reaction | 409.33753 | N |
| frame_force_balance_error | 2.5011104e-12 | N |
| frame_load_solution_valid | 1 | — |

OPTIMIZATION STUDY

# Design Baseline

Revision PA-04 900 lbf actuator selection 2026-10-27; geometry pending; axle-relative actuator offsets 2026-10-28; unified full-width inlet 2026-10-30; equal regolith densities 2026-11-01 · plate-frame frame · x44-standard motors

Named input case · design_baseline.json

2026-11-09T23:00:43+00:00

**CALCULATED route timing: BWR/motor-limited straight profiles and retained empirical point-turn equations; not hardware calibrated. Legacy traction/stability checks remain active. Deposition and relocation retain their reference timing.**

Calculated route timing: traversal_time_one_way is inactive. Speed and acceleration inputs are upper bounds. Deposition and fresh-site relocation retain reference timing.

BWR soil source-range transfer flags: 3 across the two route legs. The published soil proxy is uncalibrated for this rover; see model provenance and the BWR documentation.

Actuator offsets are measured from the drum axle: positive side is outward along the arm; positive top is above the arm. The straight tube extension is included in mass and loads. The normal bracket is assumed rigid and massless; its mass, strength and interference remain unverified.

Actuator body packaging remains unverified. New PA-04 runs enforce the selected 6/8/9/10-inch command endpoints; the body/bracket collision reserve remains diagnostic.

**NASA IPEx horizontal-force model.** Outside published source conditions; the force is a provisional transfer. Tip/feed ratio: 10.935; source condition: 8.5. Source depth labels conflict with its tables. Shaft torque still uses the existing unvalidated force-times-radius assumption. Equal/opposite drum reactions are assumed to cancel; wheel feed work covers rolling and acceleration.

**Constraints excluded from optimizer: empty_slope_traversal_margin, loaded_slope_traversal_margin, startup_position_margin.**

These checks are still calculated and included in full model feasibility. Optimizer success with a failed excluded check is not a feasible design.

## Optimizer did not converge successfully

73 / 76 physical model checks pass · 7 capabilities not assessed

Predictions are conditional on the failed checks below. These are predictions from the stated inputs, not measured performance.

**Dry rover mass**

**41.5 kg**

Before adding regolith

**Payload per cycle**

**116.4 kg**

Total carried by four drums

**Complete cycles**

**3**

Within 15 minutes; no partial delivery credit

**Delivered volume**

**209.5 L**

Complete-cycle mission estimate

## Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

**Construction points**

**2,142.49**

Weighted mass + energy points

**Mass points**

**1,532.77**

Dry-mass productivity component

**Energy points**

**609.72**

Consumed-energy productivity component

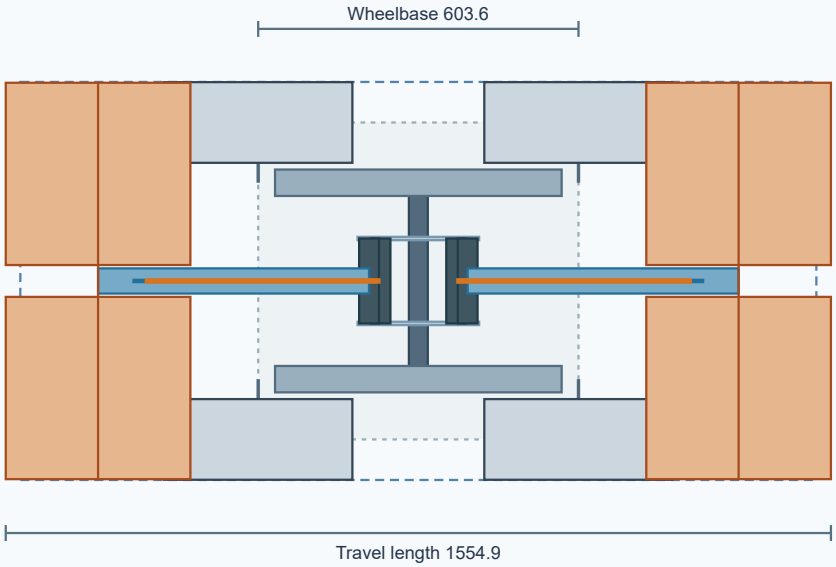
01 / ROVER CONFIGURATION

# Geometry at the evaluated inputs

Level-ground schematics and one prescribed obstacle pose. Check statuses below determine whether envelopes and clearance requirements are satisfied.

### Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.




Wheel-center track: 597.2 | Overall width: 749.2 | Wheel width: 152.0

Each drum: 347.9 diameter x 343.5 axial length | Pair gap: 60.0

Blue: one arm per drum pair · orange: actuators · charcoal: aluminum cross-tubes · slate: rockers and axle.

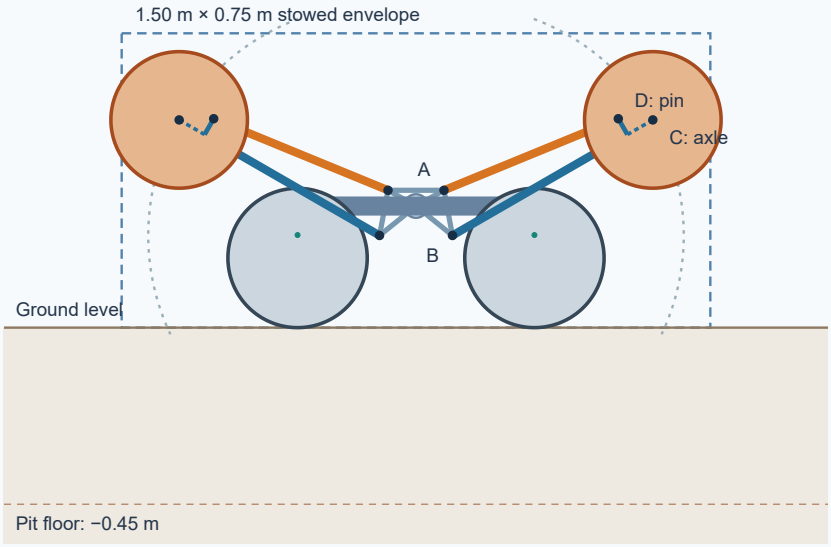
Four aluminum cross-tubes: upper/lower footprints can overlap. Dashed blue: 1500 x 750 stowed limit.

Plate outlines are schematic. Projection overlap does not prove collision. Heights and equipment are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

## Travel

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



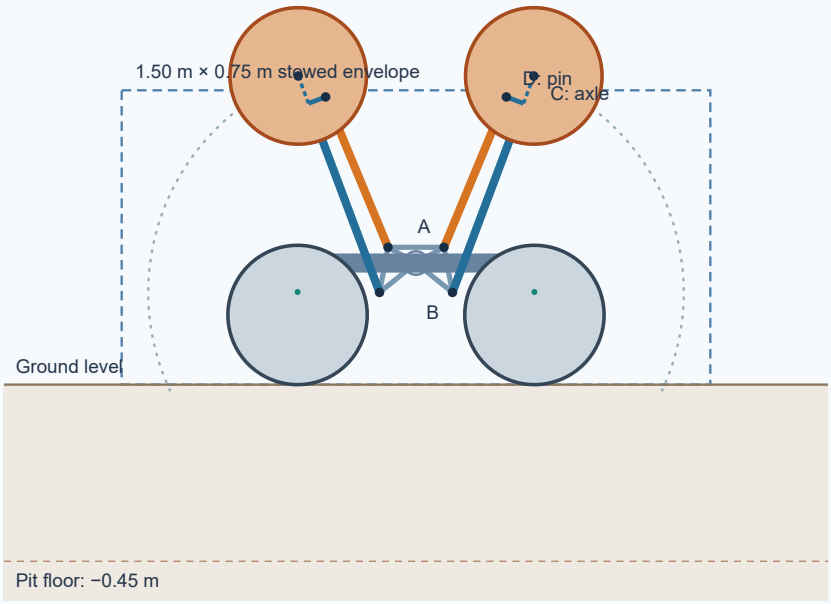

Actuator: 480.2 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.703 m · Drum bottom: +0.355 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

## Deposition · both pairs

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



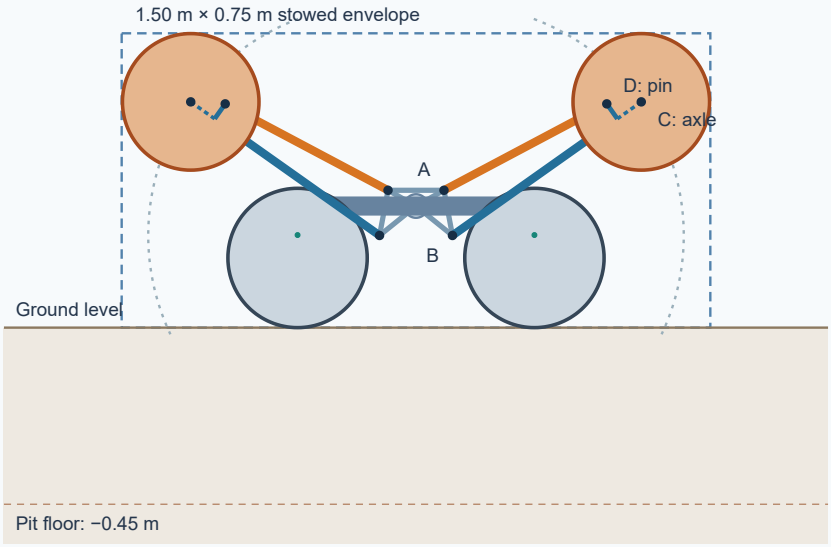

Actuator: 414.8 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.961 m · Drum bottom: +0.613 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

## Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



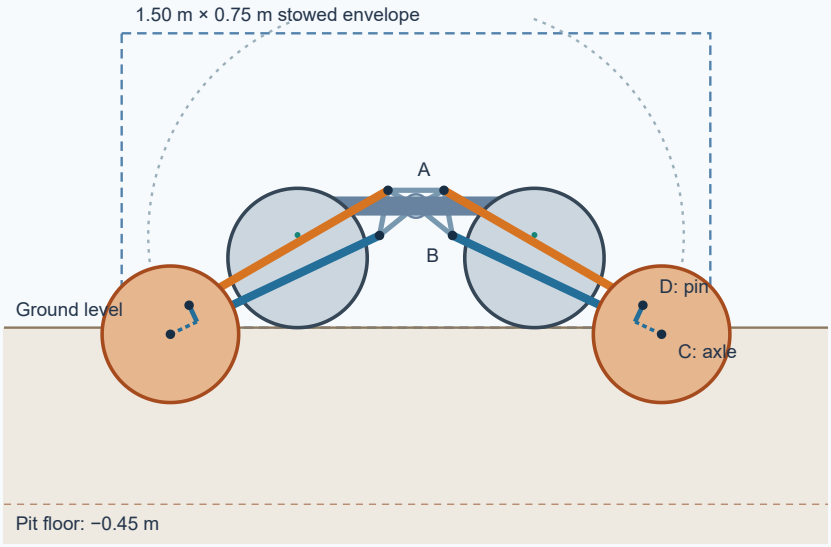

Actuator: 469.7 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.749 m · Drum bottom: +0.401 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

## Maximum extension · plunge / recovery reach

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair




Actuator: 585.5 mm · Fixed arm: 589.4 mm · Drum Ø 347.9 mm

Drum top: +0.157 m · Drum bottom: -0.191 m · A: 0.350 m · B: 0.235 m

Blue = fixed arm · Orange = actuator · Axle-to-pin offsets: side -74.7 mm / top +46.6 mm · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

## Travel and deposition mass properties

| Pose / loading | Longitudinal CG | CG height | Yaw inertia |
|---|---|---|---|
| Travel · fully loaded | 0.00 mm | 446.5 mm | 57.9 kg·m² |
| Deposition · fully loaded | 0.00 mm | 662.3 mm | 21.8 kg·m² |
| Deposition turn · one pair full | 175.2 mm | 623.6 mm | 13.2 kg·m² |
| Deposition · empty | -1.07e-14 mm | 476.2 mm | 4.55 kg·m² |

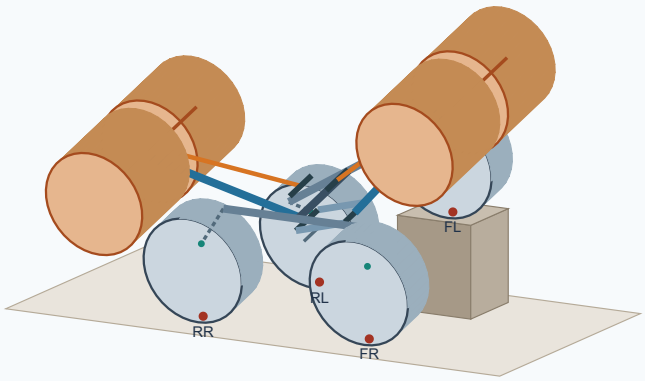
Added pose-change duration: 13.2 s. Added electrical energy: 4,349 J. Both pairs move together while full before dumping, then return empty after both dumps.

Raising reduces yaw inertia only when the drums move inward. Higher CG can reduce dynamic stability. One wheelbase must satisfy both travel and deposition checks. Half-loaded arm repositioning and dynamic dumping require controller and hardware validation.

## One wheel on the rock / travel pose

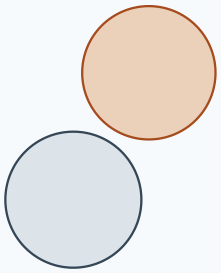
Front-left raised: 300.0 mm · chassis pitch 15.3° · roll 12.8°

Same kinematic pose as the model; arms held in travel position.




### Common roll removed

Closest cylinder pair: drum 1 / FL



Radial circle gap: +33.88 mm

Axial end gap: -150.86 mm

Finite-cylinder gap: +33.88 mm

Required clearance: 15.00 mm

One centered arm/actuator per drum pair · blue: arms · orange: actuators · charcoal: aluminum cross-tubes · gray: wheels · tan: drums.

Support block shows the prescribed height; its shape and width are illustrative. Actual rock climbing is not simulated.

Contact model omits camber effects. The right section also needs the axial-width check; circles alone can over-reject.

Parallel-cylinder clearance is the default model. This section shows the closest pair for the illustrated front-left raise.

## Plate-frame member schematic

### Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.




Plate thickness: 6.35 mm (fixed)

Common strip width: 13.32 mm

cross-tube ID / OD: 31.75 / 37.30 mm

Plate spacing / cross-tube span: 160.0 mm

Rocker bearing spacing: 370.1 mm

Wheel-to-rocker gap: 12.7 mm

Aluminum arm box: 47.43 x 48.27 mm

Arm wall: 1.0000 mm

Central support: pin, balanced mirrored gravity only.

cross-tube material properties are PROVISIONAL.

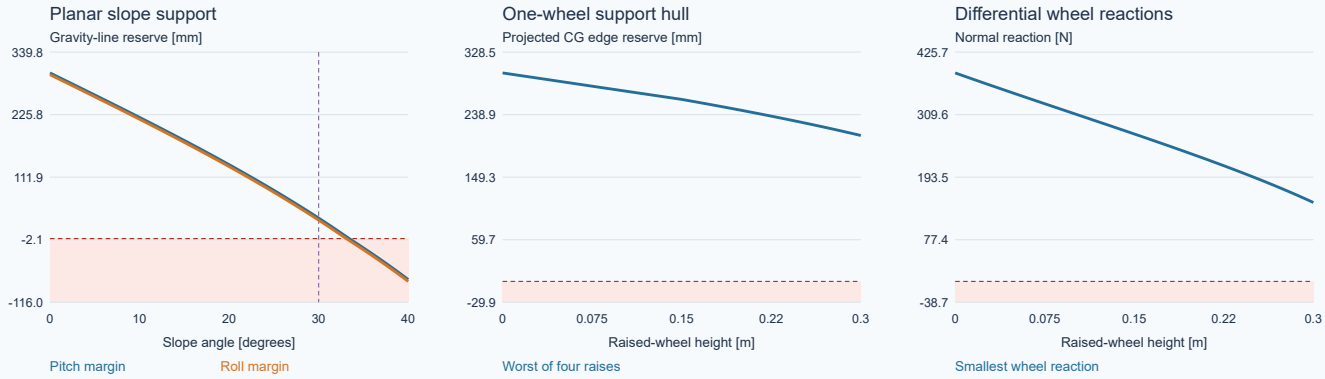
Spoke buckling is a strip screen,

not whole-plate buckling validation.

## Static slope and articulation curves

### Loaded travel pose: gravity stability screens

Linear springs + hard stops. Curves after wheel unloading are guarded continuations; slope and rock are separate cases.



LINKAGE / ACTUATOR LOADS

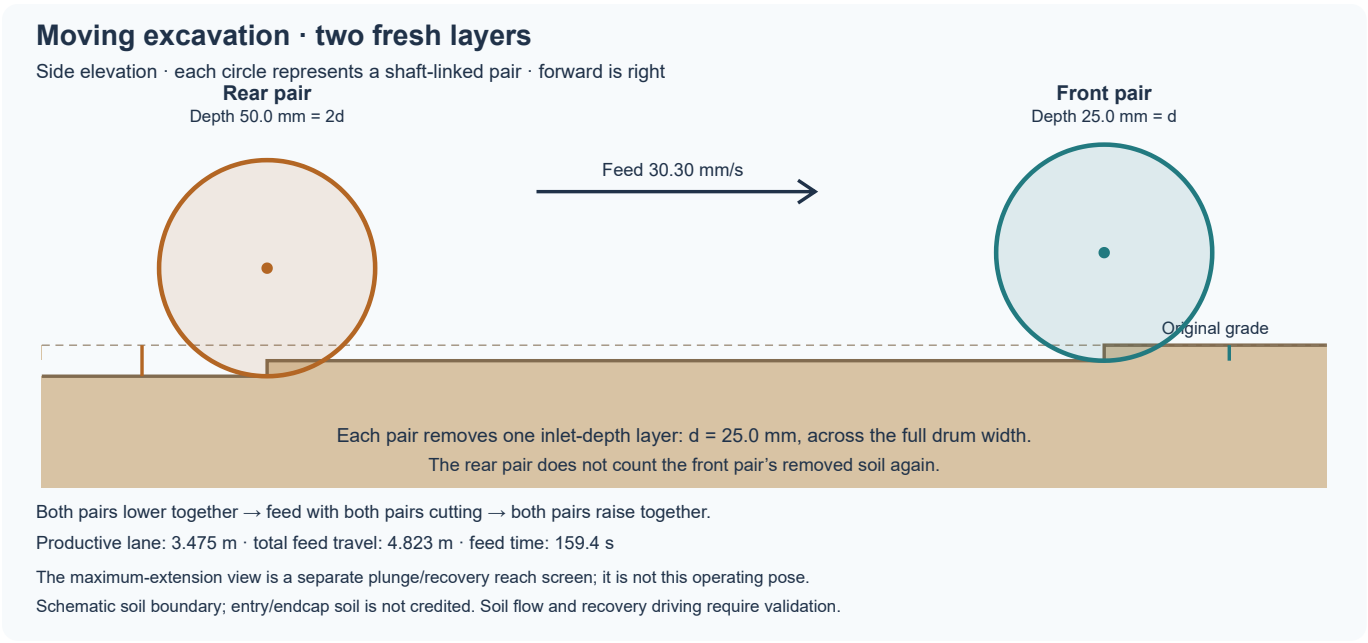
# Actuator force versus stroke



EXCAVATION / FORWARD FEED

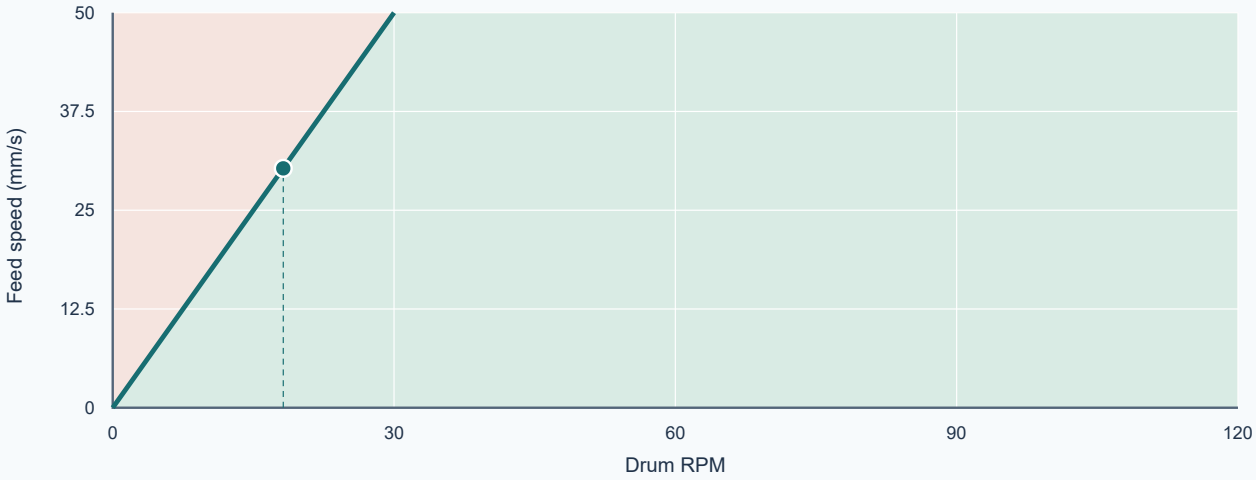
# Drive straight with the drums lowered

Each pair cuts one inlet-depth fresh layer across its full drum width. Radial protrusion equals inlet depth; the rear cuts a second layer in the same tracks. Forward advance per scoop remains limited by inlet depth.



## Feed speed and drum RPM

Forward advance per scoop ≤ inlet depth · an uncalibrated feed screen




Current: 18.19 rpm, 30.30 mm/s

Forward advance: 24.99 mm/scoop · inlet depth: 25.00 mm

Maximum feed = inlet_depth × scoops/revolution × RPM / 60; at this RPM: 30.31 mm/s.

Green: passes this feed screen. Red: excessive advance per scoop. Motor, reach and traction checks still apply.

This volume/bite constraint does not establish measured capture, cutting loads or high-speed soil retention.

| Excavation quantity | Evaluated value |
|---|---|
| Inlet depth / fresh-layer cut depth | 25.0 mm |
| Inlet width (full drum width) | 343.5 mm |
| Front depth below original grade | 25.0 mm |
| Rear depth below original grade | 50.0 mm |
| Straight excavation feed speed | 30.3 mm/s |
| Excavation drum RPM | 18.2 rpm |
| Advance per scoop | 25.0 mm |
| Feed limit at selected drum RPM | 30.3 mm/s |
| Productive excavation lane length | 3,475 mm |
| Feed travel including tandem staging | 4,823 mm |
| Straight-feed duration | 159.4 s |
| Feed-drive electrical energy | 1,537 J |
| Feed-drive peak electrical power | 13.3 W |

| Excavation quantity | Evaluated value |
|---|---|
| Plunge/recovery: axle reaches ground level | 17.0 mm |
| Lower both arms together | 8.17 s |
| Raise both arms together | 8.09 s |
| Total excavation arm motion | 16.3 s |
| Excavation arm electrical energy · both actuators | 5,229 J |

Feed-drive electrical work and peak power are included in digging totals. The old stationary-plunge soil-volume calculation is inactive. Maximum extension remains a separate capability envelope. Reaching the ground with an axle does not verify motor clearance or driving on the drums.

02 / PREDICTED OUTPUT

# Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

| Prediction | Value | Meaning |
|---|---|---|
| Full cycle duration | 290.0 s | Includes digging, actuation, travel, dumping and relocation; overlap removed once. |
| Excavation including arm motion | 183.7 s | Both arms lower together, feed/cut, then both raise together. |
| Deposition including arm motion and turn | 27.6 s | Position both arms, dump, turn, dump, and return arms to travel. |
| Excavation + deposition | 211.3 s | Excludes route, fresh-site relocation and alignment overhead. |
| Energy per cycle | 11.5 Wh | Electrical energy including auxiliaries. |
| Mission energy · complete cycles | 34.6 Wh | Includes idle electronics for remaining mission time. |
| Mission energy · continuous estimate | 35.5 Wh | Used by the existing battery constraint and MDO objective. |
| Peak electrical power | 635.7 W | Largest modeled simultaneous demand; not an average power rating. |
| Loaded mass | 157.8 kg | Dry mass + all four drums' regolith payload. |
| Traversal timing mode | calculated | BWR/motor-limited candidate profiles; legacy turn screens remain enforced. |
| Loaded outbound route | 26.0 s | Includes route turns; excludes deposition and relocation. |
| Empty return route | 26.0 s | Includes route turns; excludes deposition and relocation. |
| Actual route peak speed | 1.00 m/s | Calculated profile, not measured vehicle performance. |
| Score · complete-cycle estimate | 2,087.45 points | Uses complete deliveries and idle remainder; separate from the continuous MDO objective. |

| Prediction | Value | Meaning |
|---|---|---|
| Empty rover · 30° climb | No climbing solution | BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 30.34 N. Acceptance: solved slip strictly below 1. Soil-fit range flags: 0. |
| Full rover · 20° climb | No climbing solution | BWR, GRC-1, equal mean wheel loads. Force shortfall per wheel: 45.60 N. Acceptance: solved slip strictly below 1. Soil-fit range flags: 3. |

## Dry mass breakdown

Each assembly is counted once.

| | |
|---|---|
| Equipment allowance | **6.00 kg** |
| Frame | **4.66 kg** |
| Four bare wheels | **5.88 kg** |
| Four drive assemblies | **4.15 kg** |
| Four drums + two drive assemblies | **13.5 kg** |
| Battery | **1.58 kg** |
| Arms + actuators | **5.71 kg** |

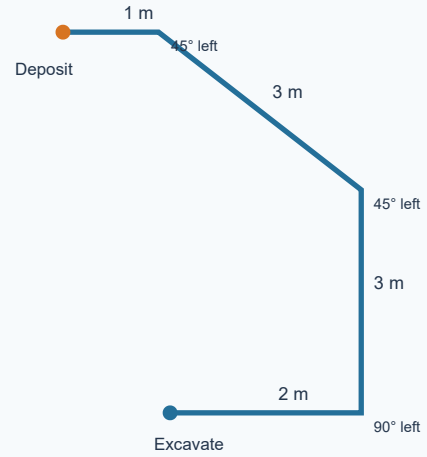
## Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

| | |
|---|---|
| Digging | **4.64 Wh** |
| Dumping | **0.15 Wh** |
| Travel + deposition turn | **1.92 Wh** |
| Excavation actuators | **1.45 Wh** |
| Deposition pose changes | **1.21 Wh** |
| Fresh-site move | **0.07 Wh** |
| Electronics | **2.01 Wh** |

## Route & cycle accounting

26.029 s outbound + 26.029 s return




Return follows the same path in reverse order; yaw signs reverse.

### 26.029 s outbound + 26.029 s return

Lower both pairs together → drive the excavation pass → raise both pairs together → outbound → move both pairs to deposition → dump one pair → turn → dump second pair → return both pairs to travel → return route → relocate to a fresh lane.

Each excavation arm phase lasts as long as the slower arm needs at its actual digging position. Lowering and cutting do not overlap. Maximum extension remains a separate load/capability check. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 290.0 s.

03 / GAPS TO RESOLVE

# Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Empty rover: 30° BWR climb (excluded from optimizer)** `empty_slope_traversal_margin` | No climbing solution `Slip = 1 is a failure sentinel` | Slip < 1 (strict) | -0.34 | Fail |
| **Full rover: 20° BWR climb (excluded from optimizer)** `loaded_slope_traversal_margin` | No climbing solution `Slip = 1 is a failure sentinel` | Slip < 1 (strict) | -0.13 | Fail |
| **Startup position margin (excluded from optimizer)** `startup_position_margin` | Reserve: -10.5 mm | Reserve ≥ 0 | -10.5 mm | Fail |
| **Dry rover mass** `dry_mass` | 41.5 kg | ≤ 80.0 kg | 38.5 kg | Pass |
| **Compact overall length** `rover_length` | 1,496 mm | ≤ 1,500 mm | 3.95 mm | Pass |
| **Overall width** `rover_width` | 749.2 mm | ≤ 750.0 mm | 0.75 mm | Pass |
| **Compact overall height** `rover_height` | 749.3 mm | ≤ 750.0 mm | 0.75 mm | Pass |
| **Ground clearance: lower axle after wheel/bar deductions** `ground_clearance_margin` | Reserve: 2.88 mm | Reserve ≥ 0 | 2.88 mm | Pass |
| **Excavation shaft-power reserve** `drum_power_margin` | Reserve: 1,189 W | Reserve ≥ 0 | 1,189 W | Pass |
| **Excavation: penetration reserve** `penetration_margin` | Reserve: 314.5 N | Reserve ≥ 0 | 314.5 N | Pass |
| **Inlet depth: radial geometry reserve** `cut_depth_margin` | Reserve: 18.5 mm | Reserve ≥ 0 | 18.5 mm | Pass |
| **Ground contact pressure** `ground_pressure_margin` | 17.9 kPa | ≤ 25.0 kPa | 7.12 kPa | Pass |
| **Straight travel: motor-force reserve** `straight_motor_margin` | Reserve: 412.6 N | Reserve ≥ 0 | 412.6 N | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Turning: motor-force reserve**<br>`turn_motor_margin` | Reserve: 5,323 N | Reserve ≥ 0 | 5,323 N | Pass |
| **Straight travel: traction reserve**<br>`straight_traction_margin` | Reserve: 169.6 N | Reserve ≥ 0 | 169.6 N | Pass |
| **Turning: traction reserve**<br>`turn_traction_margin` | Reserve: 0.30 N | Reserve ≥ 0 | 0.30 N | Pass |
| **Straight travel: shaft-power reserve**<br>`straight_power_margin` | Reserve: 2,239 W | Reserve ≥ 0 | 2,239 W | Pass |
| **Turning: shaft-power reserve**<br>`turn_power_margin` | Reserve: 2,515 W | Reserve ≥ 0 | 2,515 W | Pass |
| **Battery usable energy**<br>`battery_energy_margin` | 35.5 Wh | ≤ 176.5 Wh | 141.0 Wh | Pass |
| **Battery peak-power capacity**<br>`battery_power_margin` | 635.7 W | ≤ 1,471 W | 835.5 W | Pass |
| **Time remaining after one complete cycle**<br>`complete_cycle_margin` | Reserve: 610.0 s | Reserve ≥ 0 | 610.0 s | Pass |
| **Pit depth margin**<br>`pit_depth_margin` | Reserve: 259.1 mm | Reserve ≥ 0 | 259.1 mm | Pass |
| **Transport clearance margin**<br>`transport_clearance_margin` | Reserve: 321.3 mm | Reserve ≥ 0 | 321.3 mm | Pass |
| **Travel drum height above wheel-diameter step**<br>`wheel_first_margin` | Reserve: 0.36 mm | Reserve ≥ 0 | 0.36 mm | Pass |
| **Deposition pose: triangle reach reserve**<br>`deposition_reach_margin` | Reserve: 0.10 mm | Reserve ≥ 0 | 0.10 mm | Pass |
| **Deposition command below maximum extension**<br>`deposition_position_margin` | Reserve: 170.7 mm | Reserve ≥ 0 | 170.7 mm | Pass |
| **Deposition drum above ground**<br>`deposition_clearance_margin` | Reserve: 532.7 mm | Reserve ≥ 0 | 532.7 mm | Pass |
| **Dig position margin**<br>`dig_position_margin` | Reserve: 115.8 mm | Reserve ≥ 0 | 115.8 mm | Pass |
| **Pivot spacing margin**<br>`pivot_spacing_margin` | Reserve: 37.2 mm | Reserve ≥ 0 | 37.2 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Pivot mount margin**<br>`pivot_mount_margin` | Reserve: 355.0 mm | Reserve ≥ 0 | 355.0 mm | Pass |
| **Pivot longitudinal margin**<br>`pivot_longitudinal_margin` | Reserve: 208.9 mm | Reserve ≥ 0 | 208.9 mm | Pass |
| **Actuator pivot longitudinal margin**<br>`actuator_pivot_longitudinal_margin` | Reserve: 230.5 mm | Reserve ≥ 0 | 230.5 mm | Pass |
| **Bottom mount margin**<br>`bottom_mount_margin` | Reserve: 57.3 mm | Reserve ≥ 0 | 57.3 mm | Pass |
| **Stroke margin**<br>`stroke_margin` | Reserve: 65.3 mm | Reserve ≥ 0 | 65.3 mm | Pass |
| **Triangle retracted margin**<br>`triangle_retracted_margin` | Reserve: 65.5 mm | Reserve ≥ 0 | 65.5 mm | Pass |
| **Triangle extended margin**<br>`triangle_extended_margin` | Reserve: 33.6 mm | Reserve ≥ 0 | 33.6 mm | Pass |
| **Toggle margin**<br>`toggle_margin` | Reserve: 192.3 mm | Reserve ≥ 0 | 192.3 mm | Pass |
| **Actuator toggle margin**<br>`actuator_toggle_margin` | Reserve: 211.9 mm | Reserve ≥ 0 | 211.9 mm | Pass |
| **Wheel clearance margin**<br>`wheel_clearance_margin` | Reserve: 0.30 mm | Reserve ≥ 0 | 0.30 mm | Pass |
| **Deployed height margin**<br>`deployed_height_margin` | Reserve: 1,539 mm | Reserve ≥ 0 | 1,539 mm | Pass |
| **Actuator force capacity**<br>`actuator_force_margin` | 3,826 N | ≤ 4,003 N | 177.7 N | Pass |
| **Excavation torque reserve**<br>`drum_torque_margin` | Reserve: 260.5 N·m | Reserve ≥ 0 | 260.5 N·m | Pass |
| **Dumping torque reserve**<br>`dump_torque_margin` | Reserve: 87.5 N·m | Reserve ≥ 0 | 87.5 N·m | Pass |
| **Minimum payload reserve**<br>`payload_margin` | Reserve: 116.3 kg | Reserve ≥ 0 | 116.3 kg | Pass |
| **Excavated trench diameter reserve**<br>`trench_traverse_margin` | Reserve: 305.0 mm | Reserve ≥ 0 | 305.0 mm | Pass |
| **Fresh site margin**<br>`fresh_site_margin` | Reserve: 53.0 mm | Reserve ≥ 0 | 53.0 mm | Pass |
| **Frame section margin**<br>`frame_section_margin` | Reserve: 7.80 mm | Reserve ≥ 0 | 7.80 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Arm section margin**<br>`arm_section_margin` | Reserve: 10.9 mm | Reserve ≥ 0 | 10.9 mm | Pass |
| **Structural stress envelope**<br>`structure_stress_margin` | 119.9 MPa | ≤ 120.0 MPa | 0.10 MPa | Pass |
| **Structural deflection envelope**<br>`structure_deflection_margin` | 2.65 mm | ≤ 5.00 mm | 2.35 mm | Pass |
| **Structural buckling reserve**<br>`structure_buckling_margin` | Reserve: 2,938 N | Reserve ≥ 0 | 2,938 N | Pass |
| **Level-route dynamic tip reserve**<br>`mobility_stability_margin` | Reserve: 281.6 mm | Reserve ≥ 0 | 281.6 mm | Pass |
| **Asymmetric dump tip reserve**<br>`tip_stability_margin` | Reserve: 101.5 mm | Reserve ≥ 0 | 101.5 mm | Pass |
| **Rock pose: geometric reach reserve**<br>`obstacle_pose_margin` | Reserve: 280.6 mm | Reserve ≥ 0 | 280.6 mm | Pass |
| **Rock endpoint: global tip reserve**<br>`obstacle_tip_margin` | Reserve: 209.0 mm | Reserve ≥ 0 | 209.0 mm | Pass |
| **Obstacle drum–wheel clearance**<br>`obstacle_drum_wheel_margin` | 33.9 mm | ≥ 15.0 mm | 18.9 mm | Pass |
| **30° slope: static tip reserve**<br>`slope_tip_margin` | Reserve: 33.4 mm | Reserve ≥ 0 | 33.4 mm | Pass |
| **Rock path: global tip reserve**<br>`rock_sweep_tip_margin` | Reserve: 209.0 mm | Reserve ≥ 0 | 209.0 mm | Pass |
| **Rock path: smallest wheel reaction**<br>`rock_sweep_wheel_load_margin` | Reserve: 146.2 N | Reserve ≥ 0 | 146.2 N | Pass |
| **Available excavation area**<br>`excavation_area_margin` | Reserve: 0.01 $m^2$ | Reserve ≥ 0 | 0.01 $m^2$ | Pass |
| **Fresh-site move: drive/traction reserve**<br>`site_drive_margin` | Reserve: 156.7 N | Reserve ≥ 0 | 156.7 N | Pass |
| **Plate geometry reserve**<br>`plate_geometry_margin` | Reserve: 7.04 mm | Reserve ≥ 0 | 7.04 mm | Pass |
| **Frame packaging reserve**<br>`frame_packing_margin` | Reserve: 91.9 mm | Reserve ≥ 0 | 91.9 mm | Pass |
| **Plate strip stress**<br>`plate_stress_margin` | 96.3 MPa | ≤ 120.0 MPa | 23.7 MPa | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Plate strip buckling reserve**<br>`plate_buckling_margin` | Reserve: 2,938 N | Reserve ≥ 0 | 2,938 N | Pass |
| **aluminum cross-tube bending stress**<br>`cross_tube_bending_margin` | 94.9 MPa | ≤ 120.0 MPa | 25.1 MPa | Pass |
| **Slope: minimum normal wheel reaction**<br>`slope_wheel_load_margin` | Reserve: 0.50 N | Reserve ≥ 0 | 0.50 N | Pass |
| **Static wheel equilibrium and regular-branch reserve**<br>`wheel_equilibrium_margin` | Reserve: 0.85 | Reserve ≥ 0 | 0.85 | Pass |
| **Actuator catalog stroke domain**<br>`actuator_stroke_domain_margin` | Reserve: 152.4 mm | Reserve ≥ 0 | 152.4 mm | Pass |
| **Actuator load-curve domain**<br>`actuator_curve_margin` | Reserve: 177.7 N | Reserve ≥ 0 | 177.7 N | Pass |
| **Forward advance per scoop: inlet-depth reserve**<br>`feed_margin` | 25.0 mm | ≤ 25.0 mm | 0.01 mm | Pass |
| **Moving excavation: operating-pose reach**<br>`operating_reach_margin` | Reserve: 24.4 mm | Reserve ≥ 0 | 24.4 mm | Pass |
| **Plunge/recovery: axle reaches ground level**<br>`axle_ground_margin` | -17.0 mm | ≤ 0.00 mm | 17.0 mm | Pass |
| **Moving excavation: geometry domain**<br>`excavation_domain_margin` | Reserve: 149.0 mm | Reserve ≥ 0 | 149.0 mm | Pass |
| **Excavation feed: motor-force reserve**<br>`feed_motor_margin` | Reserve: 7,135 N | Reserve ≥ 0 | 7,135 N | Pass |
| **Excavation feed: shaft-power reserve**<br>`feed_power_margin` | Reserve: 2,669 W | Reserve ≥ 0 | 2,669 W | Pass |
| **Lin act side arm offset search range**<br>`lin_act_side_arm_offset_bounds` | -74.7 mm | -500.0 mm to -50.8 mm | 23.9 mm | Pass |
| **Lin act top arm offset search range**<br>`lin_act_top_arm_offset_bounds` | 46.6 mm | -150.0 mm to 150.0 mm | 103.4 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Drum outer diameter search range**<br>`drum_diameter_bounds` | 347.9 mm | 160.0 mm to 359.0 mm | 11.1 mm | Pass |
| **Width of each drum search range**<br>`drum_length_bounds` | 343.5 mm | 120.0 mm to 359.0 mm | 15.5 mm | Pass |
| **Excavation drum RPM search range**<br>`drum_rpm_bounds` | 18.2 rpm | 5.00 rpm to 120.0 rpm | 13.2 rpm | Pass |
| **Straight excavation feed speed search range**<br>`feed_speed_bounds` | 0.03 m/s | 0.0001 m/s to 0.50 m/s | 0.03 m/s | Pass |
| **Dumping speed search range**<br>`dump_rpm_bounds` | 59.8 rpm | 5.00 rpm to 60.0 rpm | 0.22 rpm | Pass |
| **Drum reduction ratio search range**<br>`drum_gear_ratio_bounds` | 59.7 | 9.00 to 125.0 | 50.7 | Pass |
| **Inlet depth / fresh-layer cut depth search range**<br>`inlet_depth_bounds` | 25.0 mm | 3.00 mm to 25.0 mm | 0.00 mm | Pass |
| **Arm wall thickness search range**<br>`arm_wall_bounds` | 1.00 mm | 1.00 mm to 5.00 mm | 0.00 mm | Pass |
| **Battery capacity search range**<br>`battery_capacity_bounds` | 220.7 Wh | 10.0 Wh to 300.0 Wh | 79.3 Wh | Pass |
| **Travel actuator length search range**<br>`travel_actuator_length_bounds` | 480.2 mm | 403.1 mm to 606.3 mm | 77.1 mm | Pass |
| **Deposition actuator length search range**<br>`deposition_actuator_length_bounds` | 414.8 mm | 403.1 mm to 606.3 mm | 11.7 mm | Pass |
| **Compact actuator length search range**<br>`actuator_retracted_length_bounds` | 469.7 mm | 403.1 mm to 606.3 mm | 66.6 mm | Pass |
| **Maximum-extension actuator length search range**<br>`actuator_extended_length_bounds` | 585.5 mm | 403.1 mm to 606.3 mm | 20.8 mm | Pass |
| **Actuator pivot height search range**<br>`top_pivot_height_bounds` | 350.0 mm | 350.0 mm to 700.0 mm | 0.00 mm | Pass |
| **Actuator pivot longitudinal position search range**<br>`actuator_pivot_x_bounds` | 71.3 mm | 20.0 mm to 300.0 mm | 51.3 mm | Pass |
| **Arm pivot height search range**<br>`bottom_pivot_height_bounds` | 234.8 mm | 90.0 mm to 300.0 mm | 65.2 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Arm length search range**<br>`arm_length_bounds` | 589.4 mm | 300.0 mm to 700.0 mm | 110.6 mm | Pass |
| **Arm pivot longitudinal position search range**<br>`pivot_x_bounds` | 92.9 mm | 50.0 mm to 400.0 mm | 42.9 mm | Pass |
| **Wheelbase search range**<br>`wheelbase_bounds` | 603.6 mm | 380.0 mm to 650.0 mm | 46.4 mm | Pass |
| **Wheel-center track search range**<br>`track_width_bounds` | 597.2 mm | 500.0 mm to 699.0 mm | 97.2 mm | Pass |
| **Aluminum arm box width search range**<br>`arm_box_width_bounds` | 47.4 mm | 25.0 mm to 100.0 mm | 22.4 mm | Pass |
| **Aluminum arm box height search range**<br>`arm_box_height_bounds` | 48.3 mm | 25.0 mm to 100.0 mm | 23.3 mm | Pass |
| **aluminum cross-tube span / plate spacing search range**<br>`cross_tube_span_bounds` | 160.0 mm | 160.0 mm to 460.0 mm | 0.00 mm | Pass |
| **Aluminum cross-tube wall thickness search range**<br>`cross_tube_wall_bounds` | 2.77 mm | 1.00 mm to 6.00 mm | 1.77 mm | Pass |
| **Plate strip width search range**<br>`plate_spoke_width_bounds` | 13.3 mm | 12.0 mm to 70.0 mm | 1.32 mm | Pass |
| **Rocker box width search range**<br>`rocker_box_width_bounds` | 49.7 mm | 20.0 mm to 65.0 mm | 15.3 mm | Pass |
| **Rocker box height search range**<br>`rocker_box_height_bounds` | 48.8 mm | 25.0 mm to 90.0 mm | 23.8 mm | Pass |
| **Rocker wall search range**<br>`rocker_wall_bounds` | 1.01 mm | 1.00 mm to 5.00 mm | 0.01 mm | Pass |
| **Axle outer diameter search range**<br>`axle_outer_diameter_bounds` | 35.6 mm | 25.0 mm to 90.0 mm | 10.6 mm | Pass |
| **Axle wall search range**<br>`axle_wall_bounds` | 1.11 mm | 1.00 mm to 6.00 mm | 0.11 mm | Pass |
| **Rocker pivot above wheel centers search range**<br>`rocker_pivot_offset_bounds` | 132.3 mm | 20.0 mm to 150.0 mm | 17.7 mm | Pass |
| **Wheel diameter search range**<br>`wheel_diameter_bounds` | 355.0 mm | 180.0 mm to 359.0 mm | 4.00 mm | Pass |

| Check | Prediction / reserve | Limit / criterion | Signed margin | Status |
|---|---|---|---|---|
| **Wheel width search range** `wheel_width_bounds` | 152.0 mm | 50.0 mm to 359.0 mm | 102.0 mm | Pass |
| **Drum-supported recovery driving** | Axle-to-ground reach is screened geometrically. Drum traction, motor-ground clearance, recovery loads and actual escape capability require hardware and soil tests. | — | — | Not assessed |
| **Slope motor limits / sideways sliding** | The separate BWR climb checks accept slip<1 only. Motor capability on slopes, lateral sliding and controller behavior are not evaluated. | — | — | Not assessed |
| **Real 30 cm rock traversal** | Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence. | — | — | Not assessed |
| **50 cm ditch crossing** | Ditch width/radius and contact sequence have not been defined. | — | — | Not assessed |
| **Continuous motor operation** | Stator-current limits and thermal duty remain undecided. | — | — | Not assessed |
| **Connections & material validation** | Bonded inserts, bushings, bolts, local failure and selected aluminum cross-tube properties need separate checks. | — | — | Not assessed |
| **Soil and skid-steer predictions** | Rolling, traction, capture and scrub relationships remain uncalibrated. | — | — | Not assessed |

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

04 / ENGINEERING LOADS

# Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

| Quantity / scope | Prediction | Load case / interpretation |
|---|---|---|
| Required force per actuator<br>`required_actuator_force` | 3,826 N | Quasi-static gravity envelope for loaded drums and arm; extraction resistance and dynamic loads are not modeled. |
| Frame gravity reaction at each actuator<br>`frame_gravity_actuator_force` | 5,739 N | Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load. |
| Nominal force per wheel<br>`frame_nominal_wheel_reaction` | 387.0 N | Equal-share level-ground weight. Not a rock or slope wheel-load envelope. |
| Total cutting force<br>`cut_force` | 184.4 N | Across four drums at the selected bite and engagement. |
| Worst modeled contact pressure<br>`ground_pressure` | 17.9 kPa | Route pressure estimate, including load transfer and rocker factor. |
| Structural stress envelope<br>`structure_stress` | 119.9 MPa | Envelope of local member stresses; not one assembled-frame FEA result. |
| Structural deflection envelope<br>`structure_deflection` | 2.65 mm | Conservative combination of local/relative deflections. |
| Upper/lower mount relative displacement<br>`frame_mount_relative_deflection` | 0.91 mm | Frame gravity sweep. |
| Rocker-end flexure<br>`rocker_deflection` | 0.97 mm | Local beam-screen displacement. |
| Suspension-axle flexure<br>`axle_deflection` | 0.34 mm | Axle-overhang screen. |
| Arm-tip flexure<br>`arm_deflection` | 2.65 mm | Relative arm beam displacement. |
| Loaded CG height<br>`loaded_cg_z` | 446.5 mm | Fully loaded travel pose, above level ground. |

# Model notes & engineering data

Keep assumptions and validity limits attached to the predictions.

## Obstacle and structural interpretation

Requested endpoint rock pose is geometrically reachable; this alone does not establish traversal. Static wheel reactions and body/rocker pose are solved together with linear springs and a 64.0 mm rigid compression stop. Stop contact can carry extra reaction; forces are not capped. Negative normal reactions reject the four-contact pose. The global support limit and first wheel unloading are separate diagnostics. After unloading or an invalid solve, continued geometry is guarded and is not a predicted post-lift trajectory. The heading search uses a 5-degree grid with bounded refinement; it is a numerical search, not a proof over all headings. Finite contact patches, wheel camber corrections, lateral compliance and soil sinkage are unmodeled. Wheel stiffness and the stop are student assumptions pending FEA and physical tests. Geometry/clearance schematics remain nominal.

## Obstacle drum–wheel clearance

Requested-height obstacle clearance screen. Parallel finite cylinders: shared roll is removed, then circular-plane and axial-width gaps are combined. Negative gap means overlap of the ideal solids. A common roll rotation preserves relative clearance in both models; rocker articulation still changes relative geometry.

Ditch requirement remains unverified. Static slope tipping assumes adequate friction; it does not verify climbing. The 5 mm structural limit applies to the modeled envelope of local deflections, not one assembled-frame displacement.

For the plate frame, aluminum cross-tube stiffness/strength are PROVISIONAL. Connections, dynamic obstacle loads, cutting/recovery frame cases and whole-plate buckling need separate verification.

## Standard X44 motors · assumptions

Four wheel motors at 100:1; two drum motors at 59.67:1 (exploratory gearing).

Electrical energy = output-shaft mechanical work / combined efficiency. Drive: 72%; drums: 70%. Both efficiencies are provisional.

Continuous operation unverified. Current-limit inputs of zero mean undecided. Shaft torque-speed capability is enforced separately from energy accounting.

## PA-04 actuators · 900 lbf / 12 V

Two actuators. Continuous catalog stroke: 203.2 mm; installed hardware limits: 403.1 mm to 606.3 mm. These include 0.00 mm added to each endpoint. Compact, travel, deposition and digging commands are compared with these limits diagnostically; packaging is not an active constraint.

Mass per actuator: 2.56 kg. Raising envelope: 0.01 m/s and 13.6 A; maximum-reach diagnostic times: raising 10.5 s, lowering 10.6 s. Lowering assumes rated-load speed and 14 A. Electrical energy uses 12 V × current × time for both actuators, including motor losses.

Mass fit supports 2–40 inch strokes; load curves support 0–900 lbf. Guarded out-of-domain diagnostics retain failed checks. Independent hardware, gravity-assisted lowering, duty cycle, temperature and installation-hardware mass remain unverified. A continuous stroke must be replaced with an available catalog size and reevaluated before purchase.

### Removed screens: diagnostics only

- `actuator_body_margin`: User removed this constraint; the reserve is diagnostic only and does not affect optimization or feasibility. Actual actuator packaging remains unverified.
- `dig_froude_margin`: User removed the uncalibrated speed cap; constant capture/lift response assumes no additional high-speed loss.
- `dump_froude_margin`: User removed the uncalibrated speed cap; actual dumping/retention speed response remains unverified.
- `chassis_clearance_margin`: Legacy filled rectangular envelope does not represent the open plate frame; actual frame/equipment interference remains unverified.
- `dig_traction_margin`: Inactive compatibility diagnostic. User assumes exact equal/opposite cutting reactions; excavation feed traction reserve removed.
- `dig_reach_margin`: Compatibility alias of operating_reach_margin; use the active operating_reach_margin and axle_ground_margin checks.

06 / MDO ONLY

# Optimization study

The candidate is separate from a released design baseline. Optimizer status: **False**. Saved status: failed_or_infeasible.

Continuous-score objective: 1,990.53 → 2,142.49 points. Complete-cycle estimate: 2,087.45 points.

| Changed input | Starting value | Candidate value |
|---|---|---|
| Lin act side arm offset | -50.8 mm | -74.7 mm |
| Lin act top arm offset | 60.0 mm | 46.6 mm |
| Drum outer diameter | 359.0 mm | 347.9 mm |
| Width of each drum | 340.0 mm | 343.5 mm |
| Excavation drum RPM | 30.0 rpm | 18.2 rpm |
| Straight excavation feed speed | 30.0 mm/s | 30.3 mm/s |
| Dumping speed | 60.0 rpm | 59.8 rpm |
| Drum reduction ratio | 60.0 | 59.7 |
| Arm wall thickness | 1.59 mm | 1.00 mm |
| Battery capacity | 259.2 Wh | 220.7 Wh |
| Travel actuator length | 470.0 mm | 480.2 mm |
| Deposition actuator length | 403.1 mm | 414.8 mm |
| Compact actuator length | 455.3 mm | 469.7 mm |
| Maximum-extension actuator length | 606.3 mm | 585.5 mm |
| Actuator pivot height | 370.0 mm | 350.0 mm |
| Actuator pivot longitudinal position | 80.0 mm | 71.3 mm |
| Arm pivot height | 230.0 mm | 234.8 mm |
| Arm length | 558.8 mm | 589.4 mm |
| Arm pivot longitudinal position | 110.0 mm | 92.9 mm |
| Wheelbase | 550.0 mm | 603.6 mm |
| Wheel-center track | 590.0 mm | 597.2 mm |
| Aluminum arm box width | 50.8 mm | 47.4 mm |
| Aluminum arm box height | 50.8 mm | 48.3 mm |
| Aluminum cross-tube wall thickness | 3.12 mm | 2.77 mm |
| Plate strip width | 18.8 mm | 13.3 mm |
| Rocker box width | 50.8 mm | 49.7 mm |
| Rocker box height | 50.8 mm | 48.8 mm |
| Rocker wall | 1.59 mm | 1.01 mm |
| Axle outer diameter | 33.0 mm | 35.6 mm |
| Axle wall | 1.00 mm | 1.11 mm |
| Rocker pivot above wheel centers | 96.2 mm | 132.3 mm |

APPENDIX / NATIVE UNITS

# Design inputs & output data

154 inputs supplied by file; 0 CLI overrides; 0 remaining inputs use model defaults. Legacy breakout_force is inactive; actuator loads use gravity only. Optimizer-selected design values appear in the bounds table. Frozen design variables (held fixed): wheel_diameter, wheel_width. Values below retain their native model units.

## All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `7e376f97af55b6f4e550d59b0dabdd15f9736d35e69ca89fce75fe33b0fd5e7b`.

| Input port | Effective value | Native units |
|---|---|---|
| lin_act_side_arm_offset | -0.074696458 | m |
| lin_act_top_arm_offset | 0.046602938 | m |
| drum_diameter | 0.34791832 | m |
| drum_length | 0.34348093 | m |
| drum_rpm | 18.188136 | rpm |
| feed_speed | 0.030299009 | m/s |
| dump_rpm | 59.776794 | rpm |
| drum_gear_ratio | 59.673494 | — |
| inlet_depth | 0.025 | m |
| arm_wall | 0.001 | m |
| battery_capacity | 220.66777 | W*h |
| travel_actuator_length | 0.48019992 | m |
| deposition_actuator_length | 0.41476087 | m |
| actuator_retracted_length | 0.46969352 | m |
| actuator_extended_length | 0.58547913 | m |
| top_pivot_height | 0.35 | m |
| actuator_pivot_x | 0.071287196 | m |
| bottom_pivot_height | 0.23482544 | m |
| arm_length | 0.58943709 | m |
| pivot_x | 0.09290125 | m |
| wheelbase | 0.60355609 | m |
| track_width | 0.59725 | m |
| arm_box_width | 0.047429432 | m |
| arm_box_height | 0.048274083 | m |
| cross_tube_span | 0.16 | m |
| cross_tube_wall | 0.0027741536 | m |
| plate_spoke_width | 0.013317659 | m |
| rocker_box_width | 0.049738588 | m |
| rocker_box_height | 0.048753229 | m |
| rocker_wall | 0.0010105692 | m |
| axle_outer_diameter | 0.035631502 | m |
| axle_wall | 0.0011100047 | m |

| Input port | Effective value | Native units |
|---|---|---|
| rocker_pivot_offset | 0.13234864 | m |
| minimum_ground_clearance | 0.1524 | m |
| clearance_radius_deflection_fraction | 0.2 | — |
| grouser_height | 0.018 | m |
| tipping_radius_deflection_fraction | 0.2 | — |
| tipping_reference_fill_fraction | 0.9 | — |
| wheel_compression_hard_stop | 0.064 | m |
| differential_bar_clearance_allowance | 0.0254 | m |
| empty_slope_degrees | 30 | deg |
| loaded_slope_degrees | 20 | deg |
| gravity | 9.80665 | m/s**2 |
| run_time | 900 | s |
| traversal_time_one_way | 45 | s |
| travel_speed | 1 | m/s |
| acceleration | 0.4 | m/s**2 |
| turn_angle | 3.1415927 | rad |
| overhead_time | 8 | s |
| electronics_power | 25 | W |
| actuator_idle_power | 2 | W |
| soil_density | 1500 | kg/m**3 |
| berm_density | 1500 | kg/m**3 |
| fill_fraction | 0.9 | — |
| delivery_efficiency | 0.9 | — |
| capture_fraction | 0.65 | — |
| cut_pressure | 18000 | Pa |
| internal_lift_factor | 0.15 | — |
| vertical_cut_fraction | 0.5 | — |
| force_imbalance | 0.15 | — |
| dump_revolutions | 3 | — |
| drum_efficiency | 0.7 | — |
| drum_gear_efficiency | 0.8 | — |
| drive_efficiency | 0.72 | — |
| drive_gear_efficiency | 0.8 | — |
| motor_rated_rpm | 3000 | rpm |
| traction_coefficient | 0.6 | — |
| rocker_load_factor | 1.2 | — |
| wheel_deflection_fraction | 0.07 | — |
| rolling_base | 0.04 | — |
| rolling_pressure_scale | 200000 | Pa |
| allowable_ground_pressure | 25000 | Pa |
| scrub_coefficient | 0.25 | — |
| turn_speed_fraction | 0.35 | — |
| turn_accel_fraction | 0.5 | — |
| chassis_height | 0.35 | m |
| drum_pair_gap | 0.06 | m |
| fixed_mass | 6 | kg |

| Input port | Effective value | Native units |
|---|---|---|
| wheel_areal_mass | 4 | kg/m**2 |
| drum_areal_mass | 5 | kg/m**2 |
| motor_mass_per_watt | 0.006 | kg/W |
| gear_mass_per_torque | 0.05 | kg/(N*m) |
| drive_fixed_mass | 0.35 | kg |
| drive_motor_cg_offset | 0.05842 | m |
| drum_fixed_mass | 0.5 | kg |
| battery_specific_energy | 140 | W*h/kg |
| battery_usable_fraction | 0.8 | — |
| battery_c_rate | 6.667 | 1/h |
| mass_score_weight | 1 | — |
| energy_score_weight | 1 | — |
| transport_clearance | 0.08 | m |
| mechanism_clearance | 0.015 | m |
| minimum_pivot_spacing | 0.08 | m |
| minimum_stroke | 0.04 | m |
| actuator_housing_allowance | 0.04 | m |
| pit_depth | 0.45 | m |
| actuator_efficiency | 0.65 | — |
| actuator_safety_factor | 1 | — |
| breakout_force | 100 | N |
| actuator_fixed_mass | 0.6 | kg |
| actuator_mass_per_force | 0.0004 | kg/N |
| actuator_mass_per_power | 0.003 | kg/W |
| actuator_mass_per_length | 1 | kg/m |
| stored_soil_density | 1500 | kg/m**3 |
| usable_volume_fraction | 0.9 | — |
| scoops_per_revolution | 4 | — |
| max_dig_froude | 0.3 | — |
| max_dump_froude | 0.2 | — |
| drum_spinup_time | 2 | s |
| drum_torque_service_factor | 1.25 | — |
| minimum_dig_depth | 0.005 | m |
| minimum_payload | 0.05 | kg |
| fresh_site_distance | 0.8 | m |
| site_speed | 0.1 | m/s |
| available_excavation_area | 12 | m**2 |
| tube_density | 2700 | kg/m**3 |
| tube_youngs_modulus | 6.9e+10 | Pa |
| tube_yield_strength | 2.4e+08 | Pa |
| structure_safety_factor | 2 | — |
| structure_load_factor | 1.5 | — |
| allowable_deflection | 0.005 | m |
| joint_mass | 2 | kg |
| differential_mass | 1 | kg |
| obstacle_height | 0.3 | m |

| Input port | Effective value | Native units |
|---|---|---|
| slope_angle | 0.52359878 | rad |
| frame_tube_diameter | 0.038 | m |
| frame_wall | 0.003175 | m |
| arm_tube_diameter | 0.05 | m |
| plate_thickness | 0.00635 | m |
| cross_tube_inner_diameter | 0.03175 | m |
| wheel_rocker_gap | 0.0127 | m |
| plate_hub_radius | 0.03 | m |
| plate_cutout_clearance | 0.005 | m |
| frame_mount_clearance | 0.003 | m |
| minimum_axle_overhang | 0.01 | m |
| plate_buckling_k | 1 | — |
| cross_tube_density | 2700 | kg/m**3 |
| cross_tube_youngs_modulus | 6.9e+10 | Pa |
| cross_tube_yield_strength | 2.4e+08 | Pa |
| drive_motor_power | 835.4 | W |
| drum_motor_power | 835.4 | W |
| drive_gear_ratio | 100 | — |
| motor_bus_voltage | 12 | V |
| x44_motor_mass | 0.34019428 | kg |
| drive_gearbox_mass | 0.697 | kg |
| drum_gearbox_mass | 0.753 | kg |
| drive_stator_current_limit | 0 | A |
| drum_stator_current_limit | 0 | A |
| actuator_force_rating | 4003.3995 | N |
| actuator_speed | 0.009906 | m/s |
| actuator_catalog_stroke | 0.2032 | m |
| actuator_installation_allowance | 0 | m |
| wheel_diameter | 0.355 | m |
| wheel_width | 0.152 | m |

## Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

| Port | Lower | Evaluated | Upper | Native units |
|---|---|---|---|---|
| lin_act_side_arm_offset | -0.5 | -0.0746965 | -0.0508 | m |
| lin_act_top_arm_offset | -0.15 | 0.0466029 | 0.15 | m |
| drum_diameter | 0.16 | 0.347918 | 0.359 | m |
| drum_length | 0.12 | 0.343481 | 0.359 | m |
| drum_rpm | 5 | 18.1881 | 120 | rpm |
| feed_speed | 0.0001 | 0.030299 | 0.5 | m/s |
| dump_rpm | 5 | 59.7768 | 60 | rpm |
| drum_gear_ratio | 9 | 59.6735 | 125 | — |
| inlet_depth | 0.003 | 0.025 | 0.025 | m |
| arm_wall | 0.001 | 0.001 | 0.005 | m |
| battery_capacity | 10 | 220.668 | 300 | W*h |
| travel_actuator_length | 0.403098 | 0.4802 | 0.606298 | m |
| deposition_actuator_length | 0.403098 | 0.414761 | 0.606298 | m |
| actuator_retracted_length | 0.403098 | 0.469694 | 0.606298 | m |
| actuator_extended_length | 0.403098 | 0.585479 | 0.606298 | m |
| top_pivot_height | 0.35 | 0.35 | 0.7 | m |
| actuator_pivot_x | 0.02 | 0.0712872 | 0.3 | m |
| bottom_pivot_height | 0.09 | 0.234825 | 0.3 | m |
| arm_length | 0.3 | 0.589437 | 0.7 | m |
| pivot_x | 0.05 | 0.0929013 | 0.4 | m |
| wheelbase | 0.38 | 0.603556 | 0.65 | m |
| track_width | 0.5 | 0.59725 | 0.699 | m |
| arm_box_width | 0.025 | 0.0474294 | 0.1 | m |
| arm_box_height | 0.025 | 0.0482741 | 0.1 | m |
| cross_tube_span | 0.16 | 0.16 | 0.46 | m |
| cross_tube_wall | 0.001 | 0.00277415 | 0.006 | m |
| plate_spoke_width | 0.012 | 0.0133177 | 0.07 | m |
| rocker_box_width | 0.02 | 0.0497386 | 0.065 | m |
| rocker_box_height | 0.025 | 0.0487532 | 0.09 | m |
| rocker_wall | 0.001 | 0.00101057 | 0.005 | m |
| axle_outer_diameter | 0.025 | 0.0356315 | 0.09 | m |
| axle_wall | 0.001 | 0.00111 | 0.006 | m |
| rocker_pivot_offset | 0.02 | 0.132349 | 0.15 | m |
| wheel_diameter (fixed) | 0.18 | 0.355 | 0.359 | m |
| wheel_width (fixed) | 0.05 | 0.152 | 0.359 | m |

## All output ports

| Output port | Value | Native units |
|---|---|---|
| dry_mass | 41.474134 | kg |
| payload_mass | 116.36522 | kg |
| wheel_mass | 5.8796791 | kg |
| drum_mass | 13.497812 | kg |
| drive_mass | 4.1487771 | kg |
| battery_mass | 1.5761984 | kg |
| rover_length | 1.4960501 | m |
| rover_width | 0.74925 | m |
| rover_height | 0.74925 | m |
| ground_pressure | 17881.804 | Pa |
| actual_travel_speed | 1 | m/s |
| dig_time | 167.39277 | s |
| travel_time | 52.057994 | s |
| dump_time | 10.022404 | s |
| turn_time | 4.4304517 | s |
| cycle_time | 289.95109 | s |
| cycle_energy | 11.451028 | W*h |
| cycles | 3.1039718 | — |
| berm_volume | 0.21671662 | m**3 |
| run_energy | 35.543669 | W*h |
| average_power | 142.17468 | W |
| peak_power | 635.65756 | W |
| mass_points | 1532.7676 | — |
| energy_points | 609.71933 | — |
| construction_points | 2142.487 | — |
| no_load_speed | 1.4419647 | m/s |
| loaded_motor_speed_limit | 1.0234332 | m/s |
| loaded_power_speed_limit | 5.9378282 | m/s |
| stall_drive_force | 7415.6297 | N |
| actuation_time | 16.263489 | s |
| lowering_time | 10.627823 | s |
| arm_energy | 5228.6946 | J |
| arm_peak_power | 336 | W |
| deposition_reach_margin | 9.9999971e-05 | m |
| deposition_position_margin | 0.17071827 | m |
| deposition_clearance_margin | 0.53266541 | m |
| required_actuator_force | 3825.7033 | N |
| linkage_mass | 5.7099374 | kg |
| stroke | 0.10527922 | m |
| stowed_x | 0.57406587 | m |
| stowed_y | 0.57529084 | m |
| travel_x | 0.60349906 | m |
| travel_y | 0.52931816 | m |

| Output port | Value | Native units |
|---|---|---|
| deposition_x | 0.30015422 | m |
| deposition_y | 0.78662458 | m |
| extended_x | 0.62585128 | m |
| extended_y | -0.016968825 | m |
| sweep_x_min | 0.30015422 | m |
| sweep_x_max | 0.68233834 | m |
| actuator_leverage_max | 5.9943039 | — |
| actuator_attachment_distance | 0.51684597 | m |
| arm_physical_length | 0.58943709 | m |
| arm_extension_mass | 0 | kg |
| actuator_body_margin | 0.011662867 | m |
| chassis_clearance_margin | -0.12007424 | m |
| stowed_length | 1.4960501 | m |
| stowed_height | 0.74925 | m |
| deployed_length | 1.7125619 | m |
| deployed_height | 0.96058374 | m |
| route_distance | 18 | m |
| route_turn_angle | 360 | deg |
| route_translation_time | 37.268827 | s |
| route_turn_time | 14.789167 | s |
| route_peak_acceleration | 0.4 | m/s**2 |
| deposition_actuation_time | 13.152399 | s |
| deposition_arm_energy | 4349.469 | J |
| deposition_arm_peak_power | 336 | W |
| payload_volume | 0.077576816 | m**3 |
| fresh_volume | 0.11934895 | m**3 |
| storage_capacity | 0.077576816 | m**3 |
| reachable_soil_mass | 116.36522 | kg |
| target_payload_mass | 116.36522 | kg |
| trench_depth | 0.05 | m |
| excavation_footprint | 3.8621485 | m**2 |
| cut_force | 184.41628 | N |
| drum_froude | 0.064351732 | — |
| dig_overlap_time | 0 | s |
| dig_power | 185.57422 | W |
| dump_power | 135.14218 | W |
| dig_energy | 16698.965 | J |
| dump_energy | 530.95821 | J |
| drum_power_margin | 1189.3617 | W |
| drum_torque_margin | 260.46444 | N*m |
| dump_torque_margin | 87.462976 | N*m |
| dig_froude_margin | 0.23564827 | — |
| dump_froude_margin | -0.49510291 | — |
| penetration_margin | 314.51417 | N |
| dig_traction_margin | 118.98263 | N |
| cut_depth_margin | 0.01848979 | m |

| Output port | Value | Native units |
|---|---|---|
| dig_reach_margin | 0.024361106 | m |
| payload_margin | 116.31522 | kg |
| structure_mass | 4.6617299 | kg |
| rocker_mass | 0.56862245 | kg |
| arm_linear_mass | 0.50599898 | kg/m |
| arm_envelope_radius | 0.03383762 | m |
| frame_section_margin | 0.0077978709 | m |
| arm_section_margin | 0.010857358 | m |
| structure_stress | 1.1990001e+08 | Pa |
| structure_stress_margin | 99988.852 | Pa |
| structure_deflection | 0.0026495001 | m |
| structure_deflection_margin | 0.0023504999 | m |
| structure_buckling_margin | 2937.8224 | N |
| empty_yaw_inertia | 8.7603832 | kg*m**2 |
| empty_cg_x | 5.3538094e-18 | m |
| empty_cg_z | 0.37665793 | m |
| loaded_yaw_inertia | 57.902275 | kg*m**2 |
| loaded_cg_x | 0 | m |
| loaded_cg_z | 0.44645512 | m |
| half_yaw_inertia | 33.331329 | kg*m**2 |
| half_cg_x | 0.35234094 | m |
| half_cg_z | 0.43193137 | m |
| loaded_sprung_mass | 147.24228 | kg |
| loaded_sprung_cg_x | 0 | m |
| loaded_sprung_cg_z | 0.46365473 | m |
| deposition_empty_yaw_inertia | 4.5524463 | kg*m**2 |
| deposition_empty_cg_x | -1.0707619e-17 | m |
| deposition_empty_cg_z | 0.47616316 | m |
| deposition_loaded_yaw_inertia | 21.796471 | kg*m**2 |
| deposition_loaded_cg_x | 0 | m |
| deposition_loaded_cg_z | 0.66229743 | m |
| deposition_half_yaw_inertia | 13.174459 | kg*m**2 |
| deposition_half_cg_x | 0.17523908 | m |
| deposition_half_cg_z | 0.62356569 | m |
| relocation_time | 18.631581 | s |
| relocation_energy | 256.92413 | J |
| relocation_peak_power | 20.124054 | W |
| fresh_site_margin | 0.053038135 | m |
| site_drive_margin | 156.7402 | N |
| site_turn_time | 10.381581 | s |
| mobility_energy | 6909.9147 | J |
| drive_peak_power | 610.65756 | W |
| turn_peak_power | 228.28463 | W |
| ground_pressure_margin | 7118.1963 | Pa |
| straight_motor_margin | 412.57609 | N |
| turn_motor_margin | 5322.6254 | N |

| Output port | Value | Native units |
|---|---|---|
| straight_traction_margin | 169.62584 | N |
| turn_traction_margin | 0.30000041 | N |
| straight_power_margin | 2239.3556 | W |
| turn_power_margin | 2514.6641 | W |
| mobility_stability_margin | 0.28159616 | m |
| trench_traverse_margin | 0.305 | m |
| tip_stability_margin | 0.12653896 | m |
| tip_worst_mask | 3 | — |
| obstacle_pose_margin | 0.28064423 | m |
| obstacle_tip_margin | 0.20896292 | m |
| obstacle_pose_valid | 1 | — |
| obstacle_evaluated_height | 0.3 | m |
| obstacle_worst_wheel | 0 | — |
| obstacle_chassis_pitch | 0.30566806 | rad |
| obstacle_chassis_roll | 0.24560148 | rad |
| obstacle_loaded_cg_x | -0.041193719 | m |
| obstacle_loaded_cg_y | -0.031740103 | m |
| obstacle_loaded_cg_z | 0.46103901 | m |
| obstacle_drum_wheel_gap_bound | 0.033879009 | m |
| obstacle_drum_wheel_margin | 0.018879009 | m |
| obstacle_clearance_pose_valid | 1 | — |
| obstacle_clearance_evaluated_height | 0.3 | m |
| slope_pitch_tip_margin | 0.037170472 | m |
| slope_roll_tip_margin | 0.033438545 | m |
| slope_tip_margin | 0.033438545 | m |
| slope_tip_angle | 0.57697696 | rad |
| rock_sweep_tip_margin | 0.20896292 | m |
| rock_sweep_worst_height | 0.3 | m |
| rock_sweep_raised_wheel | 0 | — |
| rock_sweep_pitch | 0.30566806 | rad |
| rock_sweep_roll | 0.24560148 | rad |
| rock_sweep_cg_x | -0.041193719 | m |
| rock_sweep_cg_y | -0.031740103 | m |
| rock_sweep_cg_z | 0.46103901 | m |
| rock_sweep_wheel_load_margin | 146.17099 | N |
| rock_sweep_load_height | 0.3 | m |
| rock_sweep_load_raised_wheel | 0 | — |
| rock_sweep_unloaded_wheel | 0 | — |
| rock_sweep_pose_valid | 1 | — |
| rock_sweep_evaluated_height | 0.3 | m |
| rock_sweep_samples | 41 | — |
| rock_sweep_force_balance_error | 2.7000624e-13 | N |
| empty_slope_slip | 1 | — |
| empty_slope_solution_valid | 0 | — |
| empty_slope_traversal_margin | -0.34454036 | — |
| empty_slope_force_shortfall | 30.339522 | N |

| Output port | Value | Native units |
|---|---|---|
| empty_slope_sinkage | 0.0062593767 | m |
| empty_slope_transfer_flag_count | 0 | — |
| loaded_slope_slip | 1 | — |
| loaded_slope_solution_valid | 0 | — |
| loaded_slope_traversal_margin | -0.12539297 | — |
| loaded_slope_force_shortfall | 45.596868 | N |
| loaded_slope_sinkage | 0.01629863 | m |
| loaded_slope_transfer_flag_count | 3 | — |
| bottom_pivot_ground_clearance | 0.15527628 | m |
| rocker_pivot_ground_clearance | 0.23113289 | m |
| ground_clearance | 0.15527628 | m |
| ground_clearance_margin | 0.0028762831 | m |
| bottom_pivot_axle_radius | 0.018649154 | m |
| rocker_pivot_axle_radius | 0.017815751 | m |
| tipping_reference_mass | 157.83936 | kg |
| tipping_wheel_outer_radius | 0.1775 | m |
| tipping_wheel_tread_radius | 0.1595 | m |
| tipping_wheel_deflection | 0.0355 | m |
| tipping_wheel_deflection_ratio | 0.2 | — |
| tipping_wheel_support_height | 0.142 | m |
| tipping_loaded_cg_z | 0.41095512 | m |
| wheel_spring_compliance | 9.1738654e-05 | m/N |
| slope_wheel_load_margin | 0.49999999 | N |
| wheel_equilibrium_margin | 0.8528906 | — |
| slope_worst_heading | 1.5707963 | rad |
| slope_load_worst_heading | 0.79966278 | rad |
| slope_contact_limit_angle | 0.52448788 | rad |
| slope_equilibrium_valid | 1 | — |
| slope_pose_error | 2.434164e-14 | m |
| rock_equilibrium_valid | 1 | — |
| rock_pose_error | 4.0592529e-16 | m |
| slope_fl_normal_load | 832.05353 | N |
| slope_fl_compression | 0.064 | m |
| slope_fl_stop_reaction | 134.41957 | N |
| slope_fr_normal_load | 246.1616 | N |
| slope_fr_compression | 0.022582534 | m |
| slope_fr_stop_reaction | 0 | N |
| slope_rl_normal_load | 261.78423 | N |
| slope_rl_compression | 0.024015733 | m |
| slope_rl_stop_reaction | 0 | N |
| slope_rr_normal_load | 0.49999999 | N |
| slope_rr_compression | 4.5869327e-05 | m |
| slope_rr_stop_reaction | 0 | N |
| obstacle_equilibrium_valid | 1 | — |
| excavation_lowering_time | 8.1685958 | s |
| excavation_raising_time | 8.0948927 | s |

| Output port | Value | Native units |
|---|---|---|
| operating_reach_margin | 0.024361106 | m |
| rear_dig_x | -0.67181813 | m |
| rear_dig_actuator_length | 0.56111803 | m |
| feed_margin | 1.1999997e-05 | m |
| front_dig_y | 0.14895916 | m |
| excavation_domain_margin | 0.14895916 | m |
| front_dig_depth | 0.025 | m |
| feed_distance | 3.4746892 | m |
| rear_dig_depth | 0.05 | m |
| inlet_width | 0.34348093 | m |
| front_dig_actuator_length | 0.55661633 | m |
| feed_per_scoop | 0.024988 | m |
| rear_dig_y | 0.12395916 | m |
| cut_depth | 0.025 | m |
| front_dig_x | 0.67605052 | m |
| feed_drive_distance | 4.8225578 | m |
| axle_ground_margin | 0.016968825 | m |
| feed_speed_limit | 0.03031356 | m/s |
| feed_time | 159.39277 | s |
| pair_cutting_time | 114.83145 | s |
| soil_supply_rate | 0.0010407132 | m**3/s |
| scoop_throughput | 0.001041213 | m**3/s |
| feed_drive_energy | 1537.2818 | J |
| feed_drive_peak_power | 13.262381 | W |
| feed_motor_margin | 7134.7594 | N |
| feed_power_margin | 2669.4801 | W |
| nasa_force_source_valid | 0 | — |
| nasa_tip_feed_ratio | 10.935425 | — |
| outbound_travel_time | 26.028997 | s |
| return_travel_time | 26.028997 | s |
| outbound_route_acceleration | 0.4 | m/s**2 |
| return_route_acceleration | 0.4 | m/s**2 |
| outbound_straight_speed_limit | 1 | m/s |
| return_straight_speed_limit | 1 | m/s |
| outbound_bwr_slip | 0.22879351 | — |
| return_bwr_slip | 0.25670922 | — |
| outbound_bwr_sinkage | 0.014469278 | m |
| return_bwr_sinkage | 0.0058184317 | m |
| outbound_bwr_transfer_flag_count | 3 | — |
| return_bwr_transfer_flag_count | 0 | — |
| pit_depth_margin | 0.25907201 | m |
| transport_clearance_margin | 0.32133168 | m |
| wheel_first_margin | 0.00035900007 | m |
| startup_position_margin | -0.010506401 | m |
| dig_position_margin | 0.11578562 | m |
| pivot_spacing_margin | 0.037185099 | m |

| Output port | Value | Native units |
|---|---|---|
| pivot_mount_margin | 0.355 | m |
| pivot_longitudinal_margin | 0.2088768 | m |
| bottom_mount_margin | 0.057325437 | m |
| stroke_margin | 0.065279216 | m |
| triangle_retracted_margin | 0.06553905 | m |
| triangle_extended_margin | 0.033551931 | m |
| toggle_margin | 0.19225297 | m |
| wheel_clearance_margin | 0.000300001 | m |
| deployed_height_margin | 1.5394163 | m |
| actuator_pivot_longitudinal_margin | 0.23049085 | m |
| actuator_toggle_margin | 0.21190561 | m |
| actuator_catalog_retracted_length | 0.403098 | m |
| actuator_catalog_extended_length | 0.606298 | m |
| actuator_unit_mass | 2.5567141 | kg |
| actuator_stroke_domain_margin | 0.1524 | m |
| actuator_force_margin | 177.6962 | N |
| actuator_curve_margin | 177.6962 | N |
| actuator_curve_valid | 1 | — |
| actuator_loaded_speed | 0.009996193 | m/s |
| actuator_raising_current | 13.556137 | A |
| actuator_lowering_current | 14 | A |
| actuator_raising_time | 10.531931 | s |
| rocker_length | 0.54005609 | m |
| plate_mass_total | 0.45287612 | kg |
| cross_tube_mass_total | 0.51993272 | kg |
| suspension_axle_mass | 0.12029859 | kg |
| suspension_bearing_span | 0.37011141 | m |
| cross_tube_outer_diameter | 0.037298307 | m |
| plate_geometry_margin | 0.0070361571 | m |
| frame_packing_margin | 0.091880706 | m |
| plate_geometry_valid | 1 | — |
| structure_cg_z | 0.30750866 | m |
| structure_sprung_cg_z | 0.30718358 | m |
| structure_yaw_inertia | 0.043659252 | kg*m**2 |
| plate_stress | 96287204 | Pa |
| plate_stress_margin | 23712796 | Pa |
| plate_buckling_margin | 2937.8224 | N |
| cross_tube_bending_stress | 94886238 | Pa |
| cross_tube_bending_margin | 25113762 | Pa |
| frame_mount_relative_deflection | 0.00090805465 | m |
| rocker_deflection | 0.00096664402 | m |
| axle_deflection | 0.00033790974 | m |
| arm_deflection | 0.002649333 | m |
| rocker_buckling_margin | 86955.955 | N |
| arm_buckling_margin | 11268.242 | N |
| frame_gravity_actuator_force | 5738.5549 | N |

| Output port | Value | Native units |
|---|---|---|
| frame_nominal_wheel_reaction | 386.96883 | N |
| frame_force_balance_error | 8.342916e-12 | N |
| frame_load_solution_valid | 1 | — |